# Retention-constrained class expansion with reference-anchored candidate tables

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**Abstract**

Class expansion must recognize arriving categories while preserving an explicit accuracy requirement on the original classes. We introduce reference-anchored candidate tables to combine a trained base classifier, semantic anchors and novel visual prototypes in a persistent score frame. Base-only pseudo-expansion episodes calibrate the mixtures, and a group offset controls novel admission. The framework measures novel recognition at a common base-accuracy constraint and the additional value of retaining the classifier across repeated arrivals. Experiments span CIFAR-100, Oxford-IIIT Pets and 200-class ImageNet-R, with ten class partitions and two frozen CLIP architectures. On ImageNet-R with ViT-L/14, the descriptive matched-retention frontier reaches 84.31% novel accuracy at 86.42% base accuracy. Across five arrivals, retaining the classifier adds 0.71 percentage points over dense prototype fusion and 1.42 points over class-mean semantic-cache retrieval in average incremental accuracy. The gain over prototypes transfers to ViT-B/32 at 0.90 points. Paired confidence intervals quantify class-partition variation on fixed data. An additional study refits all readouts with independent risk calibration.

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On ImageNet-R, the retained policy admits expansion in all ten partitions per encoder and meets every five-session test-retention target; sparse-class calibration budgets trigger explicit fallback. Raw-score and fixed-temperature controls isolate reference normalization. These results establish the predictive value of retained discrimination for repeated class expansion on ImageNet-R.
*Keywords:* Class expansion, Retention constraints, Incremental learning, Evidence fusion, Vision–language models, Long-tailed recognition

## 1. Introduction

Recognition systems often acquire new categories after deployment, while the original classifier has been fitted to imbalanced supervision. Long-tailed recognition addresses the unequal evidence available to common and rare classes [1, 2, 3]. Class expansion must put that learned discrimination to work alongside the names and sparse examples supplied for arriving categories.

Pre-trained vision–language models provide a shared interface for this evidence. CLIP maps images and descriptions into an embedding space [4]; prompt adaptation and lightweight fine-tuning specialize its predictions [5, 6, 7]; caches and prototypes incorporate labeled support [8, 9, 10]. An expanded classifier can therefore retain trained base columns and append semantic anchors and prototypes for novel classes.

The practical criterion is novel recognition under a stated base-accuracy requirement. Fixing this requirement identifies how much new-class recognition the system can admit while preserving its established service level. Successive class arrivals introduce a second criterion: the paired gain obtained by retaining the base-trained classifier throughout the sequence. Separating a

fixed-gate removal from a recalibrated classifier-free alternative measures both the contribution of the stored learned evidence and its value after calibration is restored.

We address these criteria with *reference-anchored candidate tables*. Each class stores the evidence available to it, and each branch normalizes its scores against fixed base-class references. The table retains the original learned columns and appends new rows without refitting them. This reference frame makes the existing scores independent of vocabulary size, allowing the novel admission threshold to be analyzed separately from within-group discrimination.

Class-incremental learning studies retention of earlier knowledge [11, 12], and pre-trained prototypes and analytic readouts enable efficient class addition [13, 14]. Recent methods combine multiple feature spaces [15], calibrate prototypes through sharpness-aware learning [16], and fuse CLIP prompts across modalities [17]. We quantify the retained classifier's contribution through matched-retention frontiers and paired comparisons across repeated arrivals.

On the 200-class ImageNet-R task, the retained classifier adds 0.71 [0.46, 0.92] percentage points in average incremental accuracy over dense prototype fusion with ViT-L/14 and 0.90 [0.69, 1.11] with ViT-B/32. Brackets are paired 95% bootstrap intervals across ten class partitions, conditional on the fixed image pools and encoders. Validation-selected admission policies report both novel recognition and achieved test retention. A separate refitted study adds independent protection-risk calibration and a no-admission fallback when a useful threshold cannot be certified. Removal, recalibration and replacement

controls identify the source of the sequential gain.

Our framework connects a persistent evidence table to two measurable benefits: novel recognition under a common base-accuracy requirement and the additional accuracy supplied by a retained classifier across successive arrivals. Fixed base references stabilize existing scores, and a separate admission threshold controls competition with new classes. Ten-partition comparisons with two encoders quantify these benefits against recalibration, prototype replacement and simpler normalization.

## 2. Related work

### *2.1. Imbalanced supervision and heterogeneous evidence*

Long-tailed recognition motivates balanced fitting when class evidence is unequal [2, 3]. CLIP supplies semantic anchors [4], prompt learning adapts textual evidence [5, 6], and prototypes and caches incorporate labeled support [9, 10, 8]. Our table combines these evidence types according to their availability, with a shared encoder and novel support budget across compared readouts.

### *2.2. Few-shot expansion and retained discrimination*

Multi-feature space similarity supplement (MFS3) combines base and incremental feature spaces to refine boundary decisions [15]. Sharpness-aware gradient guidance (SAGG) couples flat-region optimization with prototype calibration [16]. Adaptive exploration uses reward-guided updates to balance knowledge retention and acquisition [18]. These approaches develop representations or classifiers for successive tasks. Our formulation evaluates

the recognition capacity supplied by an existing classifier at a shared base-accuracy requirement. Fixed-gate removal, independent recalibration and dense prototype replacement identify distinct sources of its measured value.

### *2.3. Incremental recognition with pre-trained models*

Class-incremental learning evaluates a model on the union of classes observed across successive training stages. Exemplar rehearsal and representation constraints are established tools for retaining earlier knowledge [11]. Few-shot class-incremental learning adds a data scarcity condition to the arriving classes [12]. Recent work studies temporal consistency with unlabeled memory replay [19] and the geometry of inter-task and intra-task discrimination [20]. These approaches motivate explicit measurement of existing-class behavior.

SimpleCIL examines prototype classifiers built from pre-trained representations [13]; RanPAC combines random projections with accumulated class statistics [14]. Reference-anchored fusion extends this setting to semantic, trained, and support-derived evidence with different availability patterns. Calibrated seen/unseen recognition [21] provides a related perspective on the base–novel operating trade-off.

Bimodal Prompt Collaborative CLIP combines attribute-enriched text with gated visual–textual prompts for few-shot incremental learning [17]. CLIP-guided continual novel class discovery uses semi-supervised prototype adaptation to transfer visual–language knowledge to a discovery model [22]. The latter operates with unlabeled novel data; our arriving classes have names and labeled support. REAL enhances representations for exemplar-free analytic learning through complementary base pretraining and feature fusion [23].

These studies establish complementary ways to retain and combine prior knowledge. Reference-anchored tables organize semantic anchors, learned columns and support prototypes within one fixed score frame, making the retained classifier's admission value directly measurable.

FeNeC represents classes with clustered feature representatives and class-specific distances, extending nearest-class-mean recognition to local neighborhoods [24]. Its Mahalanobis and Euclidean variants provide recent frozen-feature comparisons under the same base fitting data and five-shot novel support. We report native predictions and an explicitly group-offset-calibrated form.

BiMC [25] calibrates textual classifiers with fine-grained descriptions, adjusts novel visual prototypes using base prototypes, and combines modality and covariance-based metrics. HyCal [26] combines cosine and Mahalanobis information to address heterogeneous cross-discipline few-shot streams. These methods improve the evidence or metric used for incremental recognition. Our framework instead measures the contribution of a separately fitted base-discriminative branch under a shared retention requirement and preserves an explicit score frame while heterogeneous evidence is appended. The relevant comparison is the attainable base–novel trade-off after equalizing data access and calibration, rather than an unrestricted accuracy comparison across different protocols.

TV100 studies categories unavailable during CLIP's original pre-training period [27]. Here, novelty is defined relative to task-specific classifier fitting and calibration; CLIP retains its web-scale pre-training exposure.

## 3. Expandable recognition with heterogeneous evidence

### *3.1. Problem definition*

Let $\mathcal{B}$ denote base classes with a labeled training set $\mathcal{D}_B$, and let $\mathcal{N}$ denote novel classes supplied at expansion time. The active vocabulary is initially $\mathcal{C}_0 = \mathcal{B}$ and subsequently becomes $\mathcal{C} = \mathcal{B} \cup \mathcal{N}$. An image encoder $f$ maps an input image to a normalized vector $v = f(x) \in \mathbb{R}^D$. A text encoder supplies a normalized class anchor $t_c$. For each $c \in \mathcal{N}$, the support set $\mathcal{S}_c$ contains $k$ labeled examples. Inference is over all active classes, without an oracle indicating whether an image belongs to a base or novel class.

Base training can be imbalanced. If $n_c$ is the number of training images for class $c$, the classifier may use inverse-frequency weights or another balanced fitting procedure. After this stage, the encoder, base classifier, reference banks, and branch mixing coefficients are fixed. The group admission offset is either frozen or calibrated on base validation to enforce a retention target. At arrival, novel class information is restricted to its name and $\mathcal{S}_c$. Neither previous training images nor additional labeled novel validation images are required to construct its row.

Each row has an availability mask $m_{ec} \in \{0, 1\}$ over evidence sources $e$. We use semantic evidence $T$, learned discriminative evidence $L$, and visual prototype evidence $P$. The experimental configuration is

$$(m_{Tc}, m_{Lc}, m_{Pc}) = \begin{cases} (1, 1, 0), & c \in \mathcal{B}, \\ (1, 0, 1), & c \in \mathcal{N}. \end{cases} \tag{1}$$

Base prototypes form the visual branch's normalization references. The availability mask determines which evidence contributes to a candidate; the

reference bank determines its scale.

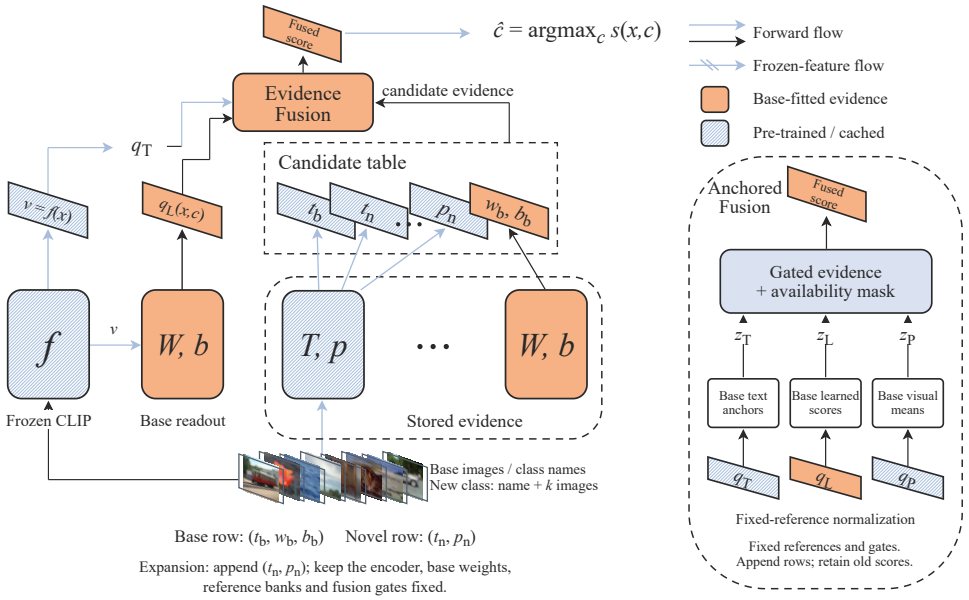



Figure 1: Reference-anchored candidate-table architecture. A shared frozen CLIP feature supplies semantic similarity and the base-trained readout. The candidate table stores text anchors for all classes, learned weights for base classes, and support prototypes for arriving classes. The inset shows the three evidence branches normalized against their respective fixed base reference sets before availability-masked fusion; $z_e = \widehat{q}_e$ denotes normalized evidence. A novel row is appended without updating the encoder, existing weights, references, or gates. Orange identifies base-fitted evidence and fusion; blue hatching identifies pre-trained or cached evidence. Branch gates remain fixed; retention control can adjust the separate novel admission offset.

### *3.2. Three notions of preservation*

Preserving a parameter vector $\theta$ means that $\theta$ receives no update during expansion. Preserving a score additionally requires

$$s_{\mathcal{C}_0}(x, c) = s_{\mathcal{C}}(x, c), \quad c \in \mathcal{C}_0. \tag{2}$$

Preserving a prediction requires the maximizing class over the larger vocabulary to remain the same. The third property is stronger than the second: a newly admitted class can have a larger score than the previous winner. We seek score preservation while allowing that competition.

The evaluation separates changes among existing classes from selection of an appended class. Under fixed evidence, reference anchoring preserves existing-class competition and admits new alternatives through their own scores.

## 4. Reference-anchored candidate tables

### *4.1. Evidence construction*

The semantic score is cosine similarity between a normalized image feature and a class text anchor,

$$q_T(x, c) = v^\top t_c. \tag{3}$$

We average normalized text embeddings of eight fixed prompts and normalize the result. The same prompts construct anchors for base and arriving classes.

The learned score is the output of a base classifier,

$$q_L(x, c) = v^\top w_c + b_c, \quad c \in \mathcal{B}. \tag{4}$$

For the frozen-encoder experiments, the weights are fitted using class-balanced ridge regression. With augmented image features $\widetilde{v} = [v; 1]$, one-hot targets $Y$, and diagonal sample-weight matrix $\Omega$, the solution is

$$W = (X^\top \Omega X + \lambda R)^{-1} X^\top \Omega Y, \tag{5}$$

where $R = \mathrm{diag}(1, \ldots, 1, 0)$ leaves the intercept unregularized. A sample from class $c$ has weight $1/n_c$, giving each base class equal total fitting weight. We solve the corresponding linear system.

The visual prototype of a novel class is

$$p_c = \frac{\sum_{x \in \mathcal{S}_c} f(x)}{\| \sum_{x \in \mathcal{S}_c} f(x) \|_2}, \qquad q_P(x, c) = v^\top p_c. \tag{6}$$

The same averaging rule applied to base training features produces a reference bank $\{p_b : b \in \mathcal{B}\}$. The novel prototype stores the information acquired at class arrival; the base bank fixes the coordinate system in which its score is interpreted.

### *4.2. Why a changing normalization population matters*

Candidate-wise standardization computes the mean and standard deviation over the current valid candidates of a branch. For text evidence, this gives

$$z_T^{\mathcal{C}}(x, c) = \frac{q_T(x, c) - \mu_T(x; \mathcal{C})}{\max(\sigma_T(x; \mathcal{C}), \epsilon)}. \tag{7}$$

Adding candidates changes $\mu_T$ and $\sigma_T$. For a text-only classifier, a common positive affine transformation preserves the ordering of the old classes. With heterogeneous fusion, however, the learned branch and text branch are rescaled differently. The relative strength of their votes can change.

A two-class example illustrates the effect. Suppose the old text scores are $(0, 2)$, giving standardized scores $(-1, 1)$, while the learned branch contributes $(0.25, -0.25)$. The second old class has the larger fused score. Appending a text score of 100 changes the two old standardized scores to approximately $(-0.728, -0.685)$. Their text separation is now about 0.043, and the unchanged learned contribution reverses the old-class ordering. No classifier weight has

changed. The example isolates a normalization effect independently of whether the new class ultimately wins the expanded decision.

*4.3. Fixed references for each evidence source*

Let $\mathcal{R}_e$ be a fixed bank of reference rows for source $e$. For every query, compute

$$\mu_e(x) = \frac{1}{|\mathcal{R}_e|} \sum_{r \in \mathcal{R}_e} q_e(x, r), \tag{8}$$

$$\sigma_e^2(x) = \frac{1}{|\mathcal{R}_e|} \sum_{r \in \mathcal{R}_e} (q_e(x, r) - \mu_e(x))^2. \tag{9}$$

The normalized candidate score is

$$\widehat{q}_e(x, c) = \frac{q_e(x, c) - \mu_e(x)}{\max(\sigma_e(x), \epsilon)}. \tag{10}$$

The reference statistics remain query-dependent, which adapts the scale to the current image, but are independent of which novel candidates are active. We use base text anchors for $\mathcal{R}_T$, base classifier columns for $\mathcal{R}_L$, and base visual means for $\mathcal{R}_P$. Each bank contains 40 reference classes on CIFAR-100 and 15 on Pets. A zero-variance reference bank uses the same fixed denominator floor $\epsilon = 10^{-6}$.

The candidate score combines available normalized evidence:

$$s(x, c) = \sum_{e \in \{T, L, P\}} m_{ec} \alpha_e \widehat{q}_e(x, c) + \delta_c. \tag{11}$$

We set $\alpha_T = 1$, write $a = \alpha_L$ and $u = \alpha_P$, and use $\delta_c = d\,\mathbf{1}[c \in \mathcal{N}]$. The resulting implementation is

$$s(x, b) = \widehat{q}_T(x, b) + a\widehat{q}_L(x, b), \quad b \in \mathcal{B}, \tag{12}$$

$$s(x, n) = \widehat{q}_T(x, n) + u\widehat{q}_P(x, n) + d, \quad n \in \mathcal{N}. \tag{13}$$

The offset $d$ calibrates competition between base and novel classes. The availability mask excludes absent learned evidence from the novel score and from reference statistics.

### *4.4. Score preservation under append-only expansion*

**Proposition 1.** *Fix an input $x$, the evidence of every existing row, the encoder, the reference banks, the denominator floor, and the coefficients in Eq.* (11)*. Appending new rows leaves the score of every existing row unchanged. Hence, every pairwise score difference and the maximizing row restricted to the existing vocabulary are unchanged, using a fixed tie-breaking rule.*

Each existing raw score, availability mask and base-reference statistic stays fixed, so their weighted sum stays fixed. The invariant applies to base and previously appended novel rows. Raw and fixed-temperature fusion share this property; the recognition comparisons in Section 5 isolate the contribution of reference normalization. Supplementary Section E provides the full proof and the consequence for expanded predictions. The implementation computes the fixed base text block separately from novel columns to keep its numerical kernel independent of the appended width.

### *4.5. Base-only calibration*

Fusion calibration must respect the distinction between base supervision and novel support. We construct a pseudo-expansion episode by splitting the base classes into approximately equal pseudo-base and pseudo-novel groups: 20/20 on CIFAR-100, 7/8 on Pets and 50/50 on ImageNet-R. A separate classifier is fitted using the pseudo-base classes alone. The pseudo-novel rows contain names and $k$ support examples from those base classes, while held-out

base validation images supply query labels. This episode simulates missing learned evidence without consuming any actual novel-class validation labels.

For each support budget, candidate coefficients are evaluated on this episode. The selection objective is the harmonic mean of macro accuracy on the two groups,

$$H = \frac{2A_B A_N}{A_B + A_N}, \tag{14}$$

We set $H = 0$ when both terms vanish. The calibration grids are $a \in \{0, 0.25, 0.5, 1, 2\}$, $u \in \{0, 0.25, 0.5, 1, 2, 4\}$, and offsets $d \in [-4, 4]$ in steps of 0.25. Ties follow a fixed grid order. The full base classifier is then combined with actual novel rows using the selected coefficients. Neither actual novel validation queries nor test labels enter this selection rule.

This procedure uses one pseudo-class split per true partition and the same support-index seed as the evaluated support condition. CIFAR and Pets pseudo splits use seed $100 + p$ for class-partition seed $p$; ImageNet-R selects half the base classes with seed $900 + p$. Coefficients are chosen once for a support budget and stay fixed when the vocabulary-size stress test appends successive subsets of novel classes.

### *4.6. Independent protection-risk calibration*

An empirical validation constraint can be supplemented with a statistical admission rule on an independent calibration pool. Let $R$ indicate a correct prediction by the retained closed-base reference and $C_d$ a correct prediction by an expanded readout at offset $d$. For retention fraction $r$, define

$$L_d = R(1 - C_d) + (1 - r)(1 - R). \tag{15}$$

Writing $D = \Pr(R = 1, C_d = 0)$, the condition $\mathbb{E}L_d \leq 1 - r$ implies $D \leq (1-r)A_{B,\mathrm{ref}}$ and hence $A_B(d) \geq rA_{B,\mathrm{ref}}$. The same sufficient criterion applies to every readout; for replacement readouts it conservatively ignores recovery of reference errors.

We adapt fixed-sequence risk testing [28] to class-stratified macro weights. Score models and the ordered threshold family are fixed using fitting and tuning data, before calibration. At $r = 0.95$, a weighted bounded-loss test examines offsets from conservative to permissive and stops at the first failed test. Each session receives error budget $\delta/5$, giving simultaneous population protection across the five prespecified arrivals for each method, order and partition under the sampling assumptions. When no finite threshold passes, the policy serves the original reference classifier with no novel admission. Supplementary Section J states the weighted concentration argument, independence requirements and fallback rule.

### *4.7. Storage and computation*

The frozen implementation uses one image feature per input. Class text anchors, base classifier weights, visual references, and novel prototypes are cached. For feature dimension $D$, the base reference banks require $O(|\mathcal{B}|D)$ storage; each novel row requires $O(D)$ additional storage for its prototype and text anchor. Once its support images have been encoded, constructing a novel prototype requires $O(kD)$ arithmetic and no optimization state.

Scoring $K$ candidates requires $O(KD)$ arithmetic in addition to the fixed reference calculations. Image encoding is shared across the vocabulary, and existing row scores can be reused when appending candidates for the same query.

## 5. Experiments

### *5.1. Data, access conditions, and implementation*

We evaluate expansion on CIFAR-100 [29] with long-tailed base supervision. The training class counts follow an exponential schedule with imbalance parameter $\rho = 100$ and nominal maximum 500 images. Five images per class are reserved before constructing the long-tailed pool, so the largest available pool contains 495 images. The remaining long-tailed pool contains 10,894 images, split into 10,487 fitting images and 407 held-out validation images. Validation reserves up to five examples per class, with smaller allocations for the rarest classes.

For seeds $\{0, \ldots, 9\}$, stratified sampling within head, medium, and tail frequency buckets yields distinct partitions of 40 base, 40 novel, and 20 unused classes. Proportional quotas use largest-remainder rounding after shuffling class membership. Evaluation uses the 80 base and novel classes.

Oxford-IIIT Pets [30] provides a complementary fine-grained task with 37 cat and dog breeds. We use its official trainval/test split and reserve five training images per breed as support. From the remaining training images, we construct an exponential long-tailed pool with $\rho = 10$ and maximum class count 90, giving 1,168 fitting and 146 validation images. Each true partition contains 15 base, 15 novel, and seven unused breeds, sampled with the same allocation rule. The official test split contains 3,669 images. Unlike CIFAR, its images have varied spatial resolutions; RGB images are resized to $224 \times 224$ by PIL bicubic interpolation before the same CLIP normalization. These features are freshly extracted for this study.

All primary expansion readouts use frozen OpenAI CLIP ViT-L/14 or

ViT-B/32 features from a single $224 \times 224$ bicubic-resized image with CLIP channel normalization. Fitting, validation, and support pools are disjoint. Supplementary Section C gives feature-cache and image-pool verification details.

Base ridge regularization is selected from $\{0.01, 0.1, 1, 10\}$ by base validation macro accuracy; the pseudo-base classifier uses the same rule. The primary comparison uses all five reserved support images, ten distinct class partitions and two arrival orders per partition. Earlier one/two/five-shot and repeated-calibration studies use three original partitions and are reported separately in Supplementary Section G.

Classifier fitting, regularization selection, and gate calibration use base classes exclusively. Five-shot CIFAR expansion consumes 200 support images for 40 novel classes. Its balanced test set supplies 4,000 base and 4,000 novel images for the expanded metrics.

ImageNet-R [31] extends the active vocabulary to 200 rendition classes: 100 base classes followed by five arrivals of 20 classes. After removing 59 byte-identical duplicates, per-class seeded splits reserve approximately 20% each for validation and test, then five support images. The fitting pool is downsampled by class rank with a nominal 100:1 schedule, yielding 2–136 images per class after availability caps. The resulting pools contain 6,438 fitting, 5,911 validation, 1,000 support and 5,911 test images. Ten frequency-stratified class partitions and two arrival orders use the same selection and reporting rules as the other datasets. RGB images are resized to $224 \times 224$ with bicubic interpolation and encoded by the same selected frozen checkpoint within each encoder comparison.

*5.2. Compared readouts and statistical unit*

Every readout shares its encoder, fitting pool, class partition and novel support. The retained table combines text and trained columns for base classes with text and support prototypes for novel classes. Fixed-gate removal sets the learned coefficient to zero; recalibrated removal independently reselects its remaining mixture. Dense anchored text–prototype fusion supplies visual means for every class. Tip-Adapter [8] supplies dense exponential semantic-cache retrieval from all permitted base fitting images and novel support. Both its published summed retrieval and an explicit class-mean control for unequal cache counts use the previously specified base-only selection grids. Recent FeNeC readouts [24] remain a three-partition controlled comparison in Supplementary Section F; their published-protocol numbers are not mixed with the expanded comparison.

Two simpler retained-classifier controls isolate normalization. Raw fusion uses cosine text/prototype scores and unnormalized ridge outputs. Fixed-temperature fusion divides each branch by one scalar, the mean per-query standard deviation across its fixed base reference columns on base-validation features. Scales are estimated once for the pseudo episode and once for the full base classifier, floored at $10^{-6}$, and remain fixed throughout arrivals. These controls use the same coefficient/offset grids and pseudo-episode objective as the retained table. All three constructions can preserve existing scores; their recognition outcomes test the benefit of per-query reference statistics.

Two arrival orders use seeds $1000 + p$ and $1100 + p$ for partition $p$. Orders are averaged within each partition. We report means and sample standard deviations over ten distinct class assignments; the original three assign-

ments are retained inside this expanded set. Paired differences use identical partition/order/session cells and, for constrained comparisons, their jointly feasible intersection. Percentile bootstrap intervals use 20,000 resamples of the paired partition means with seed 2026. These conditional intervals describe class-partition variation on shared fixed image pools and encoders, not independent-image or pretraining uncertainty. Supplementary Section I gives the full protocol and tables.

### *5.3. Validation-selected admission and achieved retention*

For retention ratio $r$, each readout receives the same absolute target $\tau_r = rA^{\text{closed}}_{B,\text{ref}}$, using the retained table's closed-base accuracy on validation. We select the largest feasible group offset from the margins of correctly classified base-validation queries, then apply it to test queries without actual novel validation or test labels. A target above a readout's within-base ceiling is infeasible. We report both test group accuracies and separately count whether $A_{B,\text{test}} \geq rA^{\text{closed}}_{B,\text{ref,test}}$. That test criterion evaluates the fixed policy; it never selects its threshold.

On ImageNet-R, all ten partitions are validation-feasible for the retained, dense-prototype and class-mean cache readouts. Their novel test accuracies are 84.63%, 83.94% and 81.61%, with base test accuracies 86.15%, 86.01% and 86.08%, respectively (Table 1). The retained policy attains the separate 95% test target in 2 of ten partitions. Validation feasibility is therefore reported independently of realized test retention. The smaller-query study in Supplementary Section H measures threshold-label demand with the original fusion calibration held fixed.

Table 1: Validation-selected 95% retention policy with ViT-L/14 after five arrivals. Accuracies are percentages, mean ± sample standard deviation over feasible class partitions. Feasible counts validation-feasible partitions; Test met counts those attaining the separate test-retention criterion. All readouts share the same ten assignments and per-partition target; available means can cover different feasible subsets. The full ViT-B/32 and eight-readout comparisons appear in Supplementary Section I.

| Data | Readout | Base | Novel | Feasible | Test met |
|---|---|---|---|---|---|
| CIFAR-100 | Retained references | 84.26 ± 1.91 | 77.10 ± 2.53 | 10/10 | 7/10 |
| CIFAR-100 | Recalibrated removal | 82.91 ± 1.62 | 66.03 ± 13.04 | 6/10 | 1/6 |
| CIFAR-100 | Dense text + prototypes | 84.00 ± 1.76 | 75.10 ± 7.15 | 8/10 | 4/8 |
| CIFAR-100 | Tip-Adapter mean | 83.19 ± 1.72 | 68.69 ± 10.39 | 9/10 | 2/9 |
| CIFAR-100 | Raw-score fusion | 82.11 ± 2.51 | 64.45 ± 13.35 | 7/10 | 1/7 |
| CIFAR-100 | Fixed temperatures | 83.72 ± 2.24 | 70.93 ± 10.56 | 10/10 | 5/10 |
| Pets | Retained references | 89.95 ± 2.67 | 93.79 ± 3.83 | 10/10 | 1/10 |
| Pets | Recalibrated removal | 90.48 ± 3.04 | 93.44 ± 3.56 | 8/10 | 3/8 |
| Pets | Dense text + prototypes | 89.46 ± 3.08 | 94.47 ± 2.60 | 10/10 | 2/10 |
| Pets | Tip-Adapter mean | 90.73 ± 2.17 | 91.39 ± 4.32 | 10/10 | 4/10 |
| Pets | Raw-score fusion | 91.35 ± 2.91 | 92.72 ± 2.70 | 9/10 | 4/9 |
| Pets | Fixed temperatures | 90.48 ± 3.42 | 94.21 ± 3.19 | 9/10 | 3/9 |
| ImageNet-R | Retained references | 86.15 ± 0.94 | 84.63 ± 0.95 | 10/10 | 2/10 |
| ImageNet-R | Recalibrated removal | 86.16 ± 0.71 | 66.88 ± 22.79 | 10/10 | 3/10 |
| ImageNet-R | Dense text + prototypes | 86.01 ± 0.94 | 83.94 ± 1.10 | 10/10 | 3/10 |
| ImageNet-R | Tip-Adapter mean | 86.08 ± 1.08 | 81.61 ± 1.51 | 10/10 | 4/10 |
| ImageNet-R | Raw-score fusion | 86.15 ± 0.68 | 65.55 ± 20.42 | 10/10 | 2/10 |
| ImageNet-R | Fixed temperatures | 85.99 ± 0.71 | 82.92 ± 1.03 | 10/10 | 1/10 |

*5.4. Independent-calibration expansion*

A separate study refits every readout after splitting the original fitting and validation pools into disjoint fitting, tuning and risk-calibration subsets, approximately 40%, 20% and 40% per class. Original support and test pools stay fixed. Regularization, mixtures, temperatures and cache settings use fitting/tuning data alone; actual novel query labels are excluded from selection. The resulting fitting budgets differ from the earlier comparisons, so the following empirical and certified policies are compared on these same newly fitted models.

The primary policy uses $r = 0.95$ and $\delta = 0.05$. Candidate offsets are 101 quantiles of base tuning margins. Fixed-sequence testing uses independent base calibration predictions and the macro-weighted protection loss in Eq. (15). The confidence statement applies to population retention across a prespecified five-session path under class-conditional independent sampling; it does not guarantee every finite test outcome or arbitrary distribution shifts. The benchmarks have been used previously in method development.

On ImageNet-R, retained tables accept finite thresholds on all ten partitions with both encoders. They achieve 89.73% base and 77.64% novel accuracy with L/14, and 73.66% base and 45.30% novel accuracy with B/32. Every final test target is met; all five sessions also meet their test targets along each of the 20 paths per encoder. Under the same sufficient protection certificate, L/14 fixed-temperature fusion admits expansion on ten partitions with 65.69% novel accuracy; dense prototypes admit five and class-mean caches two, averaging 24.12% and 6.20% novel accuracy including fallback. These are comparisons under a common protection criterion, which is conservative

Table 2: Independent-calibration policies on refitted ViT-L/14 readouts after five arrivals. Emp. uses the empirical 95% calibration constraint. Cert. exp. counts finite thresholds accepted at 95% trajectory confidence; Test met counts attainment of the separate 95% test target. Accuracies are percentages averaged over all ten partitions. Infeasible empirical policies and uncertified risk policies both fall back to the shared original reference with zero novel accuracy; fallback attainment is not useful certified expansion. All six readouts and ViT-B/32 appear in Supplementary Section J.

| Dataset | Readout | Emp. met | Emp. novel | Cert. exp. | Test met | Cert. base | Cert. novel |
|---|---|---|---|---|---|---|---|
| CIFAR-100 | Retained | 7/10 | 76.28 | 0/10 | 10/10 | 88.29 | 0.00 |
| CIFAR-100 | Dense prototypes | 6/10 | 71.92 | 0/10 | 10/10 | 88.29 | 0.00 |
| CIFAR-100 | Fixed temperature | 8/10 | 71.36 | 0/10 | 10/10 | 88.29 | 0.00 |
| CIFAR-100 | Class-mean cache | 4/10 | 65.73 | 0/10 | 10/10 | 88.29 | 0.00 |
| Pets | Retained | 1/10 | 93.07 | 0/10 | 10/10 | 96.05 | 0.00 |
| Pets | Dense prototypes | 0/10 | 95.08 | 0/10 | 10/10 | 96.05 | 0.00 |
| Pets | Fixed temperature | 3/10 | 91.87 | 0/10 | 10/10 | 96.05 | 0.00 |
| Pets | Class-mean cache | 1/10 | 92.50 | 0/10 | 10/10 | 96.05 | 0.00 |
| ImageNet-R | Retained | 7/10 | 83.45 | 10/10 | 10/10 | 89.73 | 77.64 |
| ImageNet-R | Dense prototypes | 6/10 | 82.47 | 5/10 | 10/10 | 91.04 | 24.12 |
| ImageNet-R | Fixed temperature | 5/10 | 81.28 | 10/10 | 10/10 | 89.76 | 65.69 |
| ImageNet-R | Class-mean cache | 4/10 | 79.37 | 2/10 | 10/10 | 91.18 | 6.20 |

for readouts recovering reference errors. CIFAR and Pets select fallback on all ten retained-table partitions with both encoders at this confidence level: their sparse tail-class calibration budgets do not support nontrivial certified expansion with this bound. Complete empirical safety margins and confidence levels are reported separately.

### *5.5. Classifier value over successive arrivals*

Five sessions add eight CIFAR classes, three Pets breeds or 20 ImageNet-R classes at each arrival. With the base-only selected offset fixed throughout, $\overline{A} = 5^{-1} \sum_{t=1}^{5} A_{\mathrm{all},t}$ is average incremental macro accuracy; final accuracy uses the complete active vocabulary. Table 3 reports paired retained-minus-comparator differences, separating recalibration after removal from replacement by dense prototypes or semantic caches.

On ImageNet-R, the ViT-L/14 retained table reaches 87.24% average incremental accuracy. Its paired gains are 0.71 [0.46, 0.92] points over dense prototypes and 1.42 [1.15, 1.62] over class-mean cache retrieval. With ViT-B/32, these gains are 0.90 [0.69, 1.11] and 1.71 [1.44, 1.98] points. The ImageNet-R gain therefore transfers across the two encoder architectures under identical image-pool access. CIFAR gains over classifier-free recalibration also appear with both encoders, while the dense-prototype comparison has intervals spanning zero. On Pets, average incremental intervals span zero for all three comparators with both encoders; the final B/32 comparisons can favor the alternatives. The clearest gain over prototype replacement occurs on ImageNet-R with both encoders.

Table 3: Paired classifier value with fixed base-only calibrated gates and offsets. Entries are mean gains in percentage points [conditional paired 95% bootstrap CI] across ten class partitions after averaging the two arrival orders. Positive values favor retention. Intervals resample paired class assignments on the fixed image pools.

| Encoder | Data | Comparator | Paired avg. gain [95% CI] | Paired final gain [95% CI] |
|---|---|---|---|---|
| L/14 | CIFAR-100 | Recalibrated removal | 3.00 [2.23, 3.75] | 2.62 [1.81, 3.43] |
| L/14 | CIFAR-100 | Dense text + prototypes | 0.15 [-0.46, 0.64] | -0.21 [-0.85, 0.30] |
| L/14 | CIFAR-100 | Tip-Adapter mean | 1.44 [0.68, 2.10] | 1.24 [0.41, 2.01] |
| L/14 | Pets | Recalibrated removal | -0.14 [-1.53, 0.86] | -0.78 [-2.84, 0.60] |
| L/14 | Pets | Dense text + prototypes | -0.98 [-2.52, 0.22] | -1.54 [-4.08, 0.35] |
| L/14 | Pets | Tip-Adapter mean | -0.06 [-1.76, 1.41] | -0.39 [-2.92, 1.71] |
| L/14 | ImageNet-R | Recalibrated removal | 2.30 [1.95, 2.59] | 1.71 [1.28, 2.06] |
| L/14 | ImageNet-R | Dense text + prototypes | 0.71 [0.46, 0.92] | 0.41 [0.03, 0.72] |
| L/14 | ImageNet-R | Tip-Adapter mean | 1.42 [1.15, 1.62] | 1.13 [0.76, 1.43] |
| B/32 | CIFAR-100 | Recalibrated removal | 2.38 [1.44, 3.34] | 1.80 [1.01, 2.65] |
| B/32 | CIFAR-100 | Dense text + prototypes | 0.86 [-0.15, 1.76] | 0.43 [-0.47, 1.24] |
| B/32 | CIFAR-100 | Tip-Adapter mean | 0.96 [-0.04, 1.84] | 0.51 [-0.34, 1.33] |
| B/32 | Pets | Recalibrated removal | -0.44 [-1.64, 0.81] | -2.00 [-3.80, -0.19] |
| B/32 | Pets | Dense text + prototypes | -0.39 [-1.85, 1.02] | -1.17 [-3.22, 0.67] |
| B/32 | Pets | Tip-Adapter mean | -0.74 [-1.62, 0.12] | -1.85 [-3.40, -0.59] |
| B/32 | ImageNet-R | Recalibrated removal | 3.07 [2.82, 3.34] | 2.39 [2.19, 2.61] |
| B/32 | ImageNet-R | Dense text + prototypes | 0.90 [0.69, 1.11] | 0.46 [0.26, 0.66] |
| B/32 | ImageNet-R | Tip-Adapter mean | 1.71 [1.44, 1.98] | 1.34 [1.16, 1.52] |

*5.6. Matched-test-retention capacity and simple fusion*

The descriptive frontier fixes each mixture and asks

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\text{ref,test}}^{\text{closed}}. \tag{16}$$

This sweep uses base test labels to characterize attainable capacity; it is separate from the validation-selected policy above. Sorting correctly classified base-query margins with macro weights identifies the largest feasible offset; Supplementary Section D derives its exact empirical envelope. We evaluate $r \in \{0.80, 0.85, 0.90, 0.95, 0.99\}$, with the same per-partition reference target for every readout. Infeasible targets are missing with explicit counts.

Table 4: Normalization controls with ViT-L/14. Fixed-gate average covers five arrivals; matched base/novel accuracy uses the descriptive 95% test-retention budget after the fifth arrival. Values are percentages, mean ± partition-level sample standard deviation; counts identify feasible sets. All fusion controls share the pseudo-episode grids and retained base classifier.

| Data | Fusion | Fixed-gate avg. | Matched base | Matched novel | Feasible |
|---|---|---|---|---|---|
| CIFAR-100 | Retained references | 83.03 ± 1.19 | 83.91 ± 1.42 | 78.02 ± 2.27 | 10/10 |
| CIFAR-100 | Raw-score fusion | 79.70 ± 2.64 | 83.25 ± 1.17 | 62.37 ± 6.38 | 6/10 |
| CIFAR-100 | Fixed temperatures | 81.56 ± 0.98 | 83.64 ± 1.21 | 74.53 ± 2.70 | 9/10 |
| Pets | Retained references | 92.23 ± 2.87 | 91.56 ± 1.87 | 93.30 ± 3.74 | 10/10 |
| Pets | Raw-score fusion | 87.94 ± 14.58 | 91.57 ± 1.88 | 92.39 ± 2.75 | 10/10 |
| Pets | Fixed temperatures | 89.39 ± 10.15 | 91.56 ± 1.87 | 93.64 ± 2.48 | 10/10 |
| ImageNet-R | Retained references | 87.24 ± 0.59 | 86.42 ± 0.53 | 84.31 ± 1.01 | 10/10 |
| ImageNet-R | Raw-score fusion | 83.70 ± 2.58 | 86.42 ± 0.56 | 70.85 ± 8.03 | 9/10 |
| ImageNet-R | Fixed temperatures | 86.30 ± 0.38 | 86.43 ± 0.53 | 82.49 ± 1.11 | 10/10 |

At the ImageNet-R 95% budget, the retained ViT-L/14 readout reaches 86.42% base and 84.31% novel accuracy. Relative to fixed-temperature fusion,

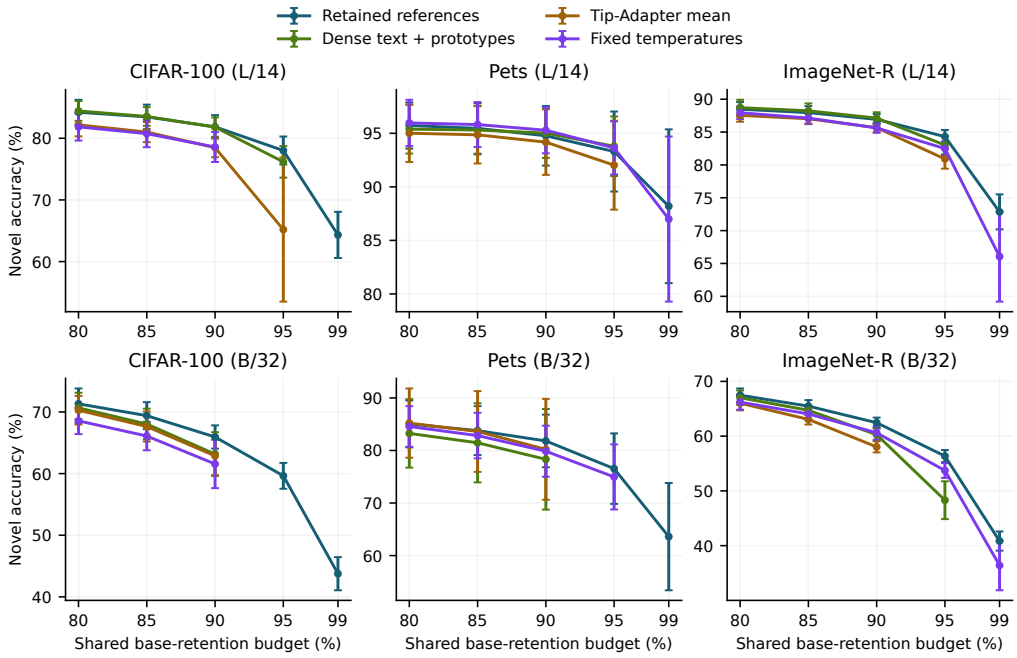



Figure 2: Descriptive novel-recognition frontiers over ten class partitions with two frozen CLIP encoders. Every readout receives the retained model's common absolute target within a partition and encoder. Error bars are partition-level sample standard deviations after averaging arrival orders. A point is drawn only when all ten partitions are feasible; Supplementary Section I reports available partial sets and counts. Thresholds use test labels for this capacity analysis and are not deployment calibration.

its paired average incremental gain is 0.94 [0.74, 1.12] points. This comparison isolates per-query branch scaling from the append-only invariant shared by both constructions. The complete budget curves characterize the admission capacity of each readout. Imbalance schedules remain fixed within each dataset, so the cross-dataset comparisons characterize operating regimes rather than the causal effect of imbalance.

## 6. Discussion

Retained discrimination has an admission value beyond its closed-base accuracy. Under a shared base requirement, a classifier that separates the original classes more effectively can admit novel candidates at a more permissive threshold. The matched frontiers express this value directly as novel accuracy, while their feasible regions identify the base service levels that each readout can support.

Sequential comparisons make the contribution measurable across repeated arrivals. Fixed-gate removal evaluates the learned branch inside a selected mixture; classifier-free recalibration measures the value that remains after restoring the mixture's operating point; dense base prototypes provide a stronger replacement alternative. These distinctions connect retained parameters to recognition outcomes rather than treating parameter preservation as the final objective.

Reference anchoring supplies a consistent score frame for this evaluation. Stored base columns and reference statistics remain fixed, branch coefficients transfer from base-only episodes, and the group threshold controls novel admission. A fixed threshold gives append-only score preservation; validation-

based threshold adjustment supplies an explicit retention policy. Both operate on the same stored classifier. Independent protection-risk calibration adds a confidence-qualified admission rule: ImageNet-R supports useful certified expansion, while the smaller tail-class calibration budgets select fallback. The statistical statement concerns population risk under the sampling assumptions; the test-attainment counts are its empirical evaluation.

## 7. Conclusion

Reference-anchored candidate tables turn a retained classifier into reusable evidence for class expansion. The framework measures novel recognition under a common base-accuracy requirement and the classifier's additional contribution over successive arrivals. On ImageNet-R, retaining learned discrimination improves average incremental accuracy over dense prototypes with both frozen encoders, while matched-retention frontiers quantify the available novel-recognition capacity. Fixed references and a separate admission threshold connect this predictive value to an expandable score system.

## CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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## Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets and ImageNet-R data and the publicly released CLIP ViT-L/14 and ViT-B/32 checkpoints. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-Tables`. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

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# Supplementary material: Retention-constrained class expansion with reference-anchored candidate tables

Junyi Zhang

Sections A–H retain the original three-partition ViT-L/14 auxiliary and validation studies. Section I reports the expanded ten-partition, two-encoder comparison; Section J adds the disjoint-pool risk-calibration study. Their statistical units are stated separately.

## Appendix A. Auxiliary readout and mechanism analyses

For the auxiliary readout and support-budget comparisons, the gate grid is $a, u \in \{0, 0.25, 0.5, 1, 2\}$ and $d \in \{-2, -1.5, \ldots, 2\}$. Offset-calibrated controls use their separately stated grids. The main retention/value evaluation uses the common expanded grid in Section 4.5. All experiments share the same frozen representations, true class partitions and disjoint support pools.

### *Appendix A.1. Readouts and evaluation measures*

All compared readouts use the same CLIP image embeddings, text anchors, class partitions, and novel support images. The text baseline uses cosine similarity to class names. The visual prototype baseline uses means of base fitting examples and means of novel support examples. Text–prototype fusion adds the two cosine branches with a mixing coefficient selected on the same pseudo-novel calibration episode; its grid is $\{0, 0.25, 0.5, 1, 2, 4\}$, including pure text.

Tip-Adapter uses a balanced $k$-image cache per candidate and scores $100q_T + \alpha q_{\text{cache}}$, with affinity $\exp[\beta(v^\top f(x_i) - 1)]$. The same base-only episode selects $\alpha \in \{0, 0.5, 1, 2, 4\}$ and $\beta \in \{1, 2, 5, 10\}$. Cache images come from the reserved pool for both groups; base-trained readouts use the full base fitting set.

We also compare raw evidence fusion and candidate-normalized fusion. Raw fusion uses unstandardized text, learned, and prototype scores with the same coefficient grid as the proposed table. Candidate-normalized fusion standardizes text over all active classes, learned scores over the base classes, and prototypes over active novel classes. It shares the base classifier and calibration protocol with reference-anchored fusion. This comparison changes the normalization population while holding the principal evidence sources fixed.

We report macro base accuracy $A_B$, macro novel accuracy $A_N$, all-class macro accuracy, and the harmonic mean in Eq. (14). Both datasets use equal numbers of base and novel classes, so all-class macro accuracy is $(A_B + A_N)/2$. On the balanced CIFAR test set this also equals sample accuracy; on Pets the macro measure gives each breed equal weight. Stability measures include maximum absolute change in an existing score, change of the maximizing class restricted to base classes, and prediction changes over the expanded vocabulary. Error flips are reported as fractions of all base test images, not conditional fractions of previously correct images.

*Appendix A.2. Recognition after five-shot class expansion*

Reference-anchored fusion reaches $86.03 \pm 1.40$ base accuracy and $66.90 \pm 4.23$ novel accuracy (Table S1). Its base accuracy is 7.71 percentage points

Table S1: Five-shot expansion from 40 base classes to 80 classes. All values are accuracy percentages, reported as mean ± sample standard deviation over three distinct class partitions. All readouts share frozen CLIP features; coefficients are selected using base-only calibration. Base and novel performance jointly specify each operating point.

| Readout | Base | Novel | All | Harmonic |
|---|---|---|---|---|
| CLIP text | 78.32 ± 0.25 | 76.86 ± 1.97 | 77.59 ± 0.86 | 77.57 ± 0.88 |
| Visual prototypes | 77.53 ± 2.03 | 60.92 ± 3.94 | 69.23 ± 2.09 | 68.17 ± 2.49 |
| Text + prototypes | 82.47 ± 0.87 | 77.53 ± 0.47 | 80.00 ± 0.56 | 79.92 ± 0.55 |
| Tip-Adapter (tuned) | 78.38 ± 0.24 | 77.42 ± 1.72 | 77.90 ± 0.83 | 77.89 ± 0.83 |
| Raw evidence fusion | 78.32 ± 0.25 | 76.86 ± 1.97 | 77.59 ± 0.86 | 77.57 ± 0.88 |
| Candidate-normalized table | 85.36 ± 0.66 | 70.83 ± 4.33 | 78.09 ± 2.02 | 77.36 ± 2.48 |
| Reference-anchored table | 86.03 ± 1.40 | 66.90 ± 4.23 | 76.46 ± 1.43 | 75.19 ± 2.14 |

above text-only classification and 3.56 points above text–prototype fusion, providing a retention-oriented operating point for class expansion.

Text–prototype fusion uses visual evidence for every class and reaches $80.00 \pm 0.56$ all-class accuracy and $77.53 \pm 0.47$ novel accuracy. Candidate-normalized fusion reaches $85.36 \pm 0.66$ base and $70.83 \pm 4.33$ novel accuracy. These points characterize the recognition trade-off accompanying the reference-anchored score system.

Raw fusion selects the text-only point in all five-shot CIFAR partitions. Both normalization rules activate learned evidence in their selected solutions, demonstrating the role of scale alignment when combining classifier logits with cosine similarities.

*Appendix A.3. Fine-grained expansion on Oxford-IIIT Pets*

On fine-grained Pets expansion, reference anchoring reaches $97.18 \pm 2.25$ base and $87.79 \pm 6.18$ novel accuracy (Table S2). Base accuracy is 2.41 points above text–prototype fusion. Mean base/novel accuracy also increases from 96.85%/86.49% with candidate normalization to 97.18%/87.79% with reference anchoring.

Table S2: Five-shot Oxford-IIIT Pets expansion from 15 to 30 breeds under long-tailed base supervision. Values are macro accuracy percentages, mean ± sample standard deviation across three distinct breed partitions. Grids and the base-only calibration rule are transferred from the CIFAR experiment without dataset-specific search expansion.

| Readout | Base | Novel | All | Harmonic |
|---|---|---|---|---|
| CLIP text | 92.87 ± 3.37 | 94.05 ± 1.35 | 93.46 ± 1.59 | 93.43 ± 1.63 |
| Visual prototypes | 92.08 ± 2.61 | 68.30 ± 4.62 | 80.19 ± 2.33 | 78.35 ± 2.97 |
| Text + prototypes | 94.77 ± 2.91 | 93.91 ± 1.55 | 94.34 ± 1.43 | 94.31 ± 1.44 |
| Tip-Adapter (tuned) | 93.56 ± 2.45 | 94.07 ± 1.18 | 93.81 ± 1.27 | 93.80 ± 1.28 |
| Raw evidence fusion | 90.13 ± 8.01 | 91.18 ± 4.27 | 90.66 ± 6.13 | 90.63 ± 6.18 |
| Candidate-normalized table | 96.85 ± 1.50 | 86.49 ± 7.64 | 91.67 ± 4.56 | 91.29 ± 4.99 |
| Reference-anchored table | 97.18 ± 2.25 | 87.79 ± 6.18 | 92.49 ± 2.83 | 92.14 ± 3.23 |

Table S3: Additional five-shot controls with the same frozen encoder, support images, and base-only calibration. Values are mean ± sample standard deviation across three partitions. The random-projection readout is a controlled frozen RanPAC-style adaptation described in the text.

CIFAR-100

| Readout | Base | Novel | All |
|---|---|---|---|
| Text + group offset | 78.93 ± 2.75 | 75.56 ± 4.40 | 77.24 ± 0.89 |
| Anchored text + prototypes | 82.40 ± 2.02 | 76.19 ± 3.18 | 79.30 ± 0.59 |
| Frozen RP-ridge (2000) | 70.18 ± 0.69 | 52.52 ± 4.27 | 61.35 ± 1.83 |

Oxford-IIIT Pets

| Readout | Base | Novel | All |
|---|---|---|---|
| Text + group offset | 95.29 ± 3.80 | 89.85 ± 4.86 | 92.57 ± 2.31 |
| Anchored text + prototypes | 94.58 ± 3.22 | 90.38 ± 5.41 | 92.48 ± 1.10 |
| Frozen RP-ridge (2000) | 93.49 ± 3.32 | 73.70 ± 5.13 | 83.60 ± 1.13 |

Text–prototype fusion reaches $94.34 \pm 1.43$ all-class macro accuracy; the reference-anchored table reaches $92.49 \pm 2.83$ with greater retention of base-breed discrimination. Both datasets use the same search grids and base-only calibration rule.

*Appendix A.4. Calibration controls and analytic incremental readouts*

Offset-calibrated controls share the same pseudo-novel episode (Table S3). Text + group offset uses base-referenced text scores and a novel offset. Anchored text + prototypes uses fixed base references for both branches and visual evidence for every class. Harmonic selection searches offsets from $-4$ to 4 in steps of 0.25 and prototype weights $\{0, 0.25, 0.5, 1, 2, 4\}$. Fixed-reference text–prototype fusion also satisfies Proposition 1; the score invariant applies

Table S4: Validation retention-targeted calibration at five shots. The pseudo-base target is 95% of the closed reference-table pseudo-base accuracy. Among feasible grid points, selection maximizes pseudo-novel accuracy; infeasible partitions use the point with highest pseudo-base accuracy, then pseudo-novel accuracy. Feasible counts refer to validation partitions. Test values are percentages, mean ± sample standard deviation; test retention is not constrained.

| Data | Readout | Base | Novel | All | Feasible |
|---|---|---|---|---|---|
| CIFAR | Text + group offset | 81.47 ± 1.13 | 71.49 ± 5.03 | 76.48 ± 1.99 | 1/3 |
| CIFAR | Anchored text + prototypes | 82.84 ± 1.50 | 77.11 ± 4.46 | 79.98 ± 1.61 | 3/3 |
| CIFAR | Candidate-normalized table | 84.69 ± 1.92 | 72.45 ± 4.14 | 78.57 ± 1.54 | 3/3 |
| CIFAR | Reference-anchored table | 84.56 ± 1.31 | 73.88 ± 5.01 | 79.22 ± 2.21 | 3/3 |
| Pets | Text + group offset | 95.29 ± 3.80 | 89.85 ± 4.86 | 92.57 ± 2.31 | 3/3 |
| Pets | Anchored text + prototypes | 94.03 ± 3.37 | 90.96 ± 5.91 | 92.50 ± 1.27 | 3/3 |
| Pets | Candidate-normalized table | 97.25 ± 1.78 | 85.26 ± 7.25 | 91.26 ± 4.40 | 3/3 |
| Pets | Reference-anchored table | 97.18 ± 2.25 | 87.79 ± 6.18 | 92.49 ± 2.83 | 3/3 |

across availability masks.

Anchored text–prototype fusion attains 82.40% base and 76.19% novel accuracy on CIFAR, and 94.58%/90.38% on Pets. Offset-calibrated text attains 78.92%/75.56% and 95.29%/89.85%, respectively. Together with the heterogeneous table, these controls separate group calibration, evidence configuration, and the shared fixed-reference property.

The analytic baseline follows the random-projection phase of RanPAC [1]. A fixed Gaussian matrix maps CLIP features to 2000 ReLU coordinates, and $(G + \lambda I)W = Q$ accumulates Gram matrices and class cross-products from base fitting data and novel support. Base validation macro accuracy selects $\lambda \in \{10^{-8}, \ldots, 10^{8}\}$; the pseudo episode selects a novel offset. This frozen adaptation uses base validation in place of the original within-task MSE selection. It attains 70.18%/52.53% base/novel accuracy on CIFAR and 93.49%/73.70% on Pets. Novel statistics refit the analytic readout; candidate-table expansion appends rows to its fixed readout.

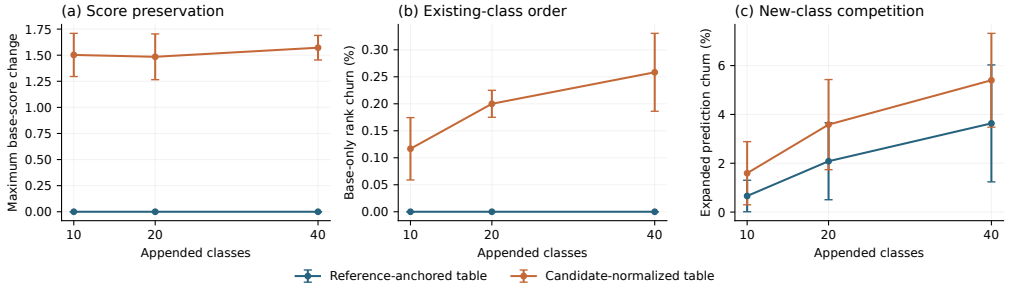



Figure S1: Vocabulary growth with five-shot novel prototypes. Each curve averages three true class partitions; error bars show sample standard deviation. Reference-anchored fusion preserves existing scores and base-only rankings. Expanded predictions can still change when a new candidate wins. Maximum score drift is measured across all test queries and existing base classes.

Under the common 95% validation retention target, reference anchoring reaches 84.56% base and 73.87% novel test accuracy on CIFAR, compared with 82.84%/77.11% for anchored text–prototype fusion (Table S4). Both are feasible on all three validation partitions. On Pets, the corresponding test points are 97.18%/87.79% and 94.03%/90.96%. Text alone reaches the CIFAR validation target in one partition. Targets constrain validation selection; the table reports the resulting test operating points. Complete 95% and 99% results accompany the experimental records.

*Appendix A.5. Vocabulary growth and score preservation*

The vocabulary test appends nested sets of 10, 20, and 40 novel classes to the 40 CIFAR base classes, using sorted novel IDs and fixed five-shot gates. At each stage, we compare existing scores with the original base decision.

Reference anchoring gives zero measured base-score drift and zero base-only ranking churn across all nine CIFAR partition and vocabulary settings (Fig. S1(a,b)). Candidate normalization changes existing scores by up to approximately 1.29–1.74 normalized units and changes a fraction of base-only

Table S5: Reference-anchored expansion across novel support budgets. Accuracy percentages are aggregated across partitions after averaging support subsets within each partition.

| Shots | Base | Novel | All | Harmonic |
|---|---|---|---|---|
| 1 | 84.57 ± 0.88 | 70.86 ± 5.41 | 77.72 ± 2.27 | 77.00 ± 2.85 |
| 2 | 85.65 ± 1.45 | 70.05 ± 1.13 | 77.85 ± 0.67 | 77.04 ± 0.67 |
| 5 | 86.03 ± 1.40 | 66.90 ± 4.23 | 76.46 ± 1.43 | 75.19 ± 2.14 |

predictions. The Pets test appends 3, 7, and 15 breeds and likewise records zero anchored score drift and ranking churn in all nine conditions.

At 40 appended classes, expanded predictions change on $3.63 \pm 2.40\%$ of base images, including $1.84 \pm 1.54\%$ correct-to-wrong changes (Fig. S1(c)). Wrong-to-correct changes are zero, as specified by Corollary 1. The preserved base scores isolate these changes as competition from appended rows.

Five-session expansion adds eight CIFAR classes or three Pets breeds per session, with two random arrival orders per partition. Across 30 session records per dataset, comparisons of all previously admitted rows, including novel rows, give maximum floating-point score differences of $2.85 \times 10^{-5}$ on CIFAR and $3.58 \times 10^{-6}$ on Pets. Both arrival orders produce identical final predictions in every partition under class-ID tie breaking.

*Appendix A.6. Support budget and evidence availability*

At one, two, and five shots, base accuracy remains between 84.57% and 86.03%, while novel accuracy ranges from 66.90% to 70.86% (Table S5; Fig. S2). Calibration includes zero branch coefficients, allowing the selected mixture to adapt to the support budget. Each budget defines a separately calibrated operating point: support determines the prototype evidence, and fusion determines its competition with learned base discrimination.

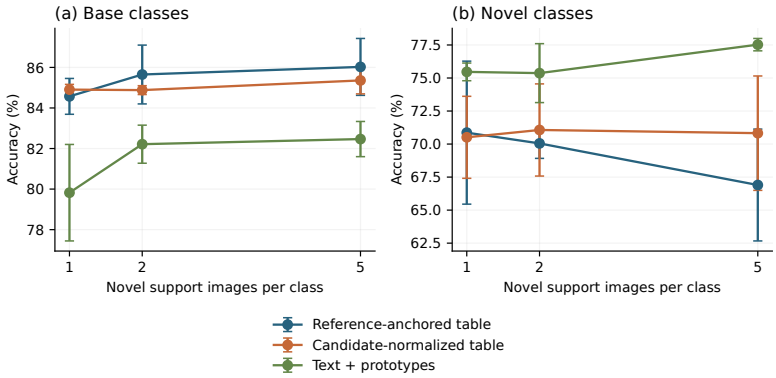



Figure S2: Base and novel accuracy at one, two, and five support images per novel class. The same fixed-reference scoring rule is used at every budget, with coefficients selected on the corresponding base-only pseudo episode. Error bars represent variation between class partitions.

Table S6: Evidence removal at the selected five-shot reference-anchored operating point, with all remaining coefficients fixed. Accuracy percentages are mean ± sample standard deviation over three class partitions.

| Readout | Base | Novel | All | Harmonic |
|---|---|---|---|---|
| Reference-anchored table | 86.03 ± 1.40 | 66.90 ± 4.23 | 76.46 ± 1.43 | 75.19 ± 2.14 |
| Without learned evidence | 58.10 ± 32.21 | 80.72 ± 4.47 | 69.41 ± 13.87 | 62.85 ± 25.08 |
| Without novel prototypes | 86.42 ± 1.71 | 47.17 ± 30.85 | 66.79 ± 14.62 | 56.15 ± 30.05 |
| Without offset | 86.14 ± 1.69 | 65.58 ± 7.35 | 75.86 ± 2.83 | 74.25 ± 4.08 |

### *Appendix A.7. Mechanism ablation and operating trade-off*

Removing learned evidence, novel prototypes, or the offset changes a different part of the selected decision rule (Table S6). With the remaining coefficients fixed, these interventions identify how each source contributes to the five-shot operating point. The independently calibrated text and text–prototype readouts provide the corresponding reduced-model comparisons.

Figure S3 sweeps $d$ from $-2$ to 2 in steps of 0.5 with $a$ and $u$ fixed. Increasing $d$ shifts the margin between the best base and novel candidates while preserving within-group rankings (Eq. (13)). The resulting curve identifies

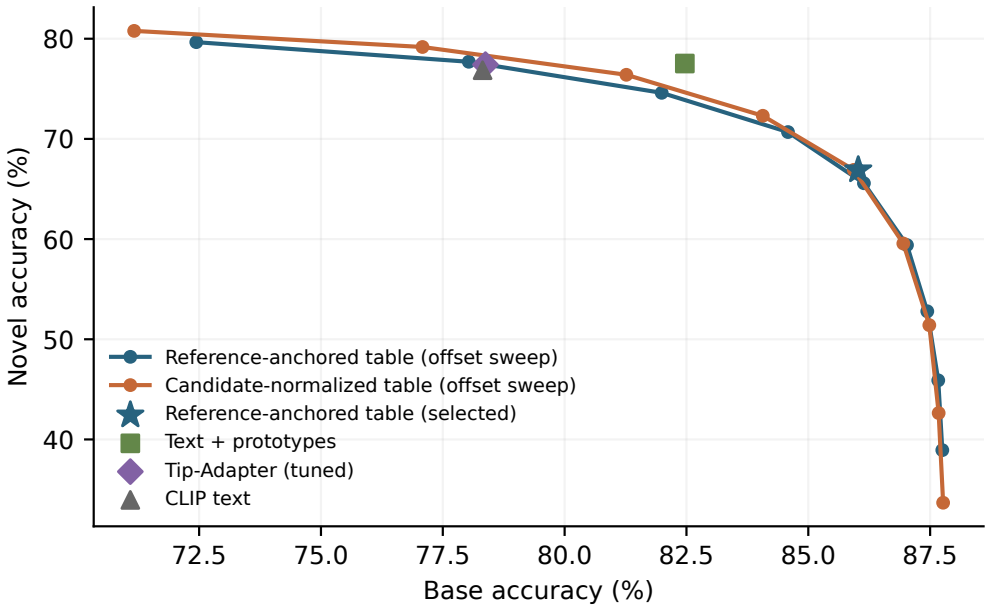



Figure S3: Base–novel operating curves for five-shot expansion. The offset is swept with the other selected coefficients fixed; these test curves are descriptive analyses and are not used to select the reported operating points. Symbols identify base-only calibrated points and matched baselines.

how the offset controls admission of new categories; a validation retention target selects the operating point.

*Appendix A.8. Readout efficiency*

Candidate scoring from cached 768-dimensional features takes approximately 1.76, 0.75, and 0.78 ms per batch for 50, 60, and 80 candidates. Measurements use batches of 256 on an NVIDIA RTX 4070 Laptop GPU, five warm-up iterations, and 50 timed iterations with GPU synchronization. They include reference statistics and fusion and exclude image/text encoding. At these vocabulary sizes, launch and reduction overheads dominate the measured latency.

Table S7: Final-session test accuracies after the admission threshold is selected using base validation only. Budgets use the same closed-base validation reference for all methods within a partition. Values are percentages, mean ± sample standard deviation over feasible partitions; counts identify those partitions. Test accuracies are the outcomes of this operational policy and are not matched by test labels.

| Data | Budget | Readout | Base | Novel | Feasible |
|---|---|---|---|---|---|
| CIFAR | 90% | Retained classifier | 78.33 ± 3.89 | 80.13 ± 2.25 | 3/3 |
| CIFAR | 90% | Classifier removed (fixed gates) | 79.07 ± 3.94 | 70.79 ± 13.81 | 3/3 |
| CIFAR | 90% | No classifier (recalibrated) | 78.73 ± 3.21 | 75.45 ± 5.43 | 3/3 |
| CIFAR | 90% | Anchored text + prototypes | 80.18 ± 0.60 | 79.45 ± 0.76 | 3/3 |
| CIFAR | 95% | Retained classifier | 82.99 ± 2.29 | 75.71 ± 3.14 | 3/3 |
| CIFAR | 95% | Classifier removed (fixed gates) | 81.95 ± 0.04 | 71.54 ± 3.87 | 2/3 |
| CIFAR | 95% | No classifier (recalibrated) | 81.85 ± 0.18 | 71.70 ± 4.10 | 2/3 |
| CIFAR | 95% | Anchored text + prototypes | 81.95 ± 0.64 | 77.48 ± 1.52 | 2/3 |
| CIFAR | 99% | Retained classifier | 86.48 ± 1.65 | 62.83 ± 10.11 | 3/3 |
| CIFAR | 99% | Classifier removed (fixed gates) | – | – | 0/3 |
| CIFAR | 99% | No classifier (recalibrated) | – | – | 0/3 |
| CIFAR | 99% | Anchored text + prototypes | 86.85 | 53.85 | 1/3 |
| Pets | 90% | Retained classifier | 84.87 ± 4.59 | 96.53 ± 0.78 | 3/3 |
| Pets | 90% | Classifier removed (fixed gates) | 84.11 ± 2.76 | 96.44 ± 0.64 | 3/3 |
| Pets | 90% | No classifier (recalibrated) | 82.00 ± 0.20 | 96.38 ± 0.52 | 3/3 |
| Pets | 90% | Anchored text + prototypes | 85.69 ± 4.62 | 95.87 ± 1.57 | 3/3 |
| Pets | 95% | Retained classifier | 90.29 ± 2.90 | 95.68 ± 0.83 | 3/3 |
| Pets | 95% | Classifier removed (fixed gates) | 90.11 ± 1.17 | 95.71 ± 0.11 | 2/3 |
| Pets | 95% | No classifier (recalibrated) | 89.88 ± 0.85 | 95.71 ± 0.11 | 2/3 |
| Pets | 95% | Anchored text + prototypes | 89.37 ± 4.15 | 95.63 ± 1.55 | 3/3 |
| Pets | 99% | Retained classifier | 96.43 ± 2.15 | 88.67 ± 7.89 | 3/3 |
| Pets | 99% | Classifier removed (fixed gates) | – | – | 0/3 |
| Pets | 99% | No classifier (recalibrated) | – | – | 0/3 |
| Pets | 99% | Anchored text + prototypes | 97.70 ± 1.11 | 84.19 ± 12.46 | 2/3 |

## Appendix B. Validation-selected retention policy

Table S7 applies the common validation target after five arrivals. Raw records include all five target fractions and every arrival session. Infeasible targets are reported without substituting a zero novel accuracy. Comparisons with different feasible counts refer to different partition sets.

## Appendix C. Reproduction details

The expansion experiment is implemented in `src/experiments/fcs_expansion.py`. It can be run from any working directory. The manifest records the code hash, frozen-cache hash, image-pool hashes, class assignments, pseudo-class assignments, support indices, regularization choices, and selection grids. The frozen support features are computed with the same deterministic interpolation used for the frozen training cache. The cached image embeddings are checked numerically before scoring.

Every novel support image comes from the five-image-per-class reserve created before long-tail sampling. This supplies a consistent support budget even for tail classes whose fitting pool contains fewer than five images. At five shots the three subset seeds select the same images in different orders, so the reported standard deviation is calculated across the three distinct class partitions. One- and two-shot results average the three subsets within each partition before computing the partition-level variation.

The main result JSON contains all compared readouts, including evidence-removal ablations. The vocabulary records contain the maximum score drift and both restricted and expanded prediction churn. The operating-curve records contain every swept offset, with no selection based on test outcomes. The efficiency records specify batch size and measured vocabulary size. The manuscript tables are generated directly from these files, and the summary CSV exposes both means and sample standard deviations.

The semantic prompts are “a photo of a {}.”, “a bad photo of a {}.”, “a cropped photo of a {}.”, “a close-up photo of a {}.”, “a bright photo of a {}.”, “a good photo of a {}.”, “a photo of the {}.”, and a second occurrence of “a

photo of a {}.". The repeated prompt follows the existing cached anchor construction and is retained consistently for all methods.

Frozen CLIP features are the sole image representation in the expansion experiments. Six fitting images are independently re-embedded and checked for cosine agreement above 0.999 with the cached features. Image hashes verify that fitting, validation, and reserved-support pools are disjoint. The fixed base text block is computed separately from novel text columns; sequential prototype products can exhibit the reported floating-point residuals when matrix widths change.

## Appendix D. Interpreting group competition

Let $M_B(x) = \max_{b \in \mathcal{B}}[\widehat{q}_T(x, b) + a\widehat{q}_L(x, b)]$ and $M_N(x) = \max_{n \in \mathcal{N}}[\widehat{q}_T(x, n) + u\widehat{q}_P(x, n)]$. The classifier selects a novel class precisely when $M_N(x) + d > M_B(x)$, subject to the fixed tie-breaking rule. Hence, the offset is a threshold on the margin $M_B - M_N$. Increasing $d$ cannot change the winning base class or the winning novel class when all other quantities are fixed. It changes which group supplies the final winner. This explains the direction of the operating curves without assuming a calibrated probability interpretation for the fused score.

For nested candidate sets and fixed references, adding novel rows can increase $M_N$ but cannot change $M_B$. An existing-class example may therefore cross the admission boundary as the vocabulary grows, while its base scores remain identical. This distinguishes the deployment behavior measured by expanded prediction churn from the normalization behavior measured by base-only ranking churn.

*Appendix D.1. Exact empirical retention envelope*

For a fixed readout, let $b_i$ and $n_i$ be the canonical winning base and novel class for query $i$, and let $\Delta_i = M_B(x_i) - M_N(x_i)$. Define $g_i(d) = \mathbf{1}[d > \Delta_i \text{ or } (d = \Delta_i \text{ and } n_i < b_i)]$. With the class-macro weights $w_i$, the two group accuracies are

$$A_B(d) = \sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i](1 - g_i(d)), \tag{D.1}$$

$$A_N(d) = \sum_{i:y_i \in \mathcal{N}} w_i \mathbf{1}[n_i = y_i] g_i(d). \tag{D.2}$$

Thus $A_B$ is nonincreasing and $A_N$ is nondecreasing in $d$. For the stored finite-precision scores, an exact search need only consider each observed margin and its immediately preceding representable offset, together with the all-base and all-novel endpoints. Between adjacent margins every group decision is fixed. If $\tau$ exceeds $\sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i]$, the constraint is infeasible. Otherwise, sorting the margins of correctly classified base queries in descending order and accumulating their macro weights locates the boundary at which retained base mass first reaches $\tau$. The tie rule selects that boundary or its predecessor. This is the largest feasible offset in the implementation's offset ordering. Monotonicity of $A_N$ proves that it maximizes novel accuracy under the empirical constraint. The argument applies identically to base-validation selection; that policy's test outcome is evaluated on separate queries.

The envelope separates the within-base feasible ceiling, the admitted group threshold and the distribution of correctly classified novel-query margins. It therefore measures the joint operating value of a learned readout and its competition with arriving evidence, beyond the statement that stored scores are preserved.

## Appendix E. Additional background and score-preservation proof

### *Appendix E.1. Long-tailed recognition and calibration*

Long-tailed recognition addresses the bias induced by unequal class frequencies. Margin adjustment allocates different classification margins to frequent and rare classes [2]; class-balanced weighting uses effective sample counts [3]; and prior-aware objectives modify the relationship between logits and class probabilities [4, 5]. Representation–classifier decoupling provides a complementary route: a representation is learned first and a balanced classifier is then fitted to that representation [6]. These developments motivate our use of a balanced base classifier as one source of evidence.

Foundation models change the representation-learning budget and the role of task-specific adaptation. LIFT studies lightweight adaptation of pretrained models for long-tail learning [7]. Low-rank updates provide an efficient mechanism for adapting large parameter matrices [8], including transformer backbones [9]. Our candidate interface can accept weights obtained through these routes. The expansion experiments here fix the encoder throughout and isolate the effect of the readout.

Long-tailed calibration studies confidence under class imbalance [10], while robust long-tailed learning addresses its interaction with label noise [11]. We study the reference population used to standardize evidence as candidate classes change. Class-balanced validation and separate base/novel metrics expose how the resulting score system allocates discrimination.

### *Appendix E.2. Semantic and visual evidence for few-shot recognition*

CLIP constructs classifiers from natural-language supervision and supports classification by changing the text candidates [12]. CoOp learns prompt

context for target tasks, while CoCoOp introduces conditional prompting to improve transfer across categories [13, 14]. These methods concern the construction or adaptation of semantic evidence. Our table keeps semantic anchors available to every class and combines them with evidence that may exist for only a subset of the vocabulary.

Prototype methods summarize class examples in an embedding space [15]. SimpleShot studies feature normalization for nearest-neighbor classification [16], and Tip-Adapter augments CLIP with a support-feature cache [17]. Few-shot transfer under distribution shift further motivates separating shared and task-specific knowledge [18]. Our comparisons use a shared encoder and novel support budget to examine evidence availability and its normalization reference.

*Appendix E.3. Proof of the main score-preservation proposition*

*Proof.* For an existing candidate $c$, each raw score $q_e(x, c)$ and mask $m_{ec}$ is fixed. The means and variances in Eq. (9) are functions of $x$ and $\mathcal{R}_e$, and neither depends on the appended rows. Therefore each normalized score in Eq. (10) is unchanged. The coefficients and offset of $c$ are fixed, so their sum is unchanged. Subtracting the unchanged scores of any two existing rows gives an unchanged difference. A fixed tie-breaking rule preserves the maximizing existing row. □

The proposition covers both base rows and previously appended novel rows with fixed prototypes. The implementation computes the fixed base text block separately from novel columns to keep its numerical kernel independent of the appended width.

**Corollary 1.** *If every existing row has a preserved score, a changed final prediction after expansion must select an appended row. An example whose previous prediction was incorrect cannot become correct after adding classes when its true class belongs to the existing vocabulary.*

*Proof.* The best existing row is unchanged by Proposition 1. The expanded maximizer is either this row or a newly appended row. For an example labeled by an existing class, selecting a new row cannot correct a previous error, and keeping the existing maximizer leaves the previous error unchanged. □

## Appendix F. Recent readout comparison and ImageNet-R extension

The extension protocol was recorded before ImageNet-R evaluation. Class partitions use seeds 0, 1 and 2, with exact frequency-stratified allocation of 100 base and 100 novel classes. Within each base group, seed $900 + p$ assigns 50 pseudo-base and 50 pseudo-novel classes. Base-only validation selects ridge regularization, branch mixtures and readout hyperparameters. No ImageNet-R test result selects these quantities. The retained table, removal controls and dense prototype replacement use the same grids as the CIFAR/Pets experiment.

FeNeC [19] is independently implemented from the paper's equations and checked against author-code commit `1575ea019fd583149c47c068f7f5ccc102ea4580`. Both variants use identity Tukey transformation for signed CLIP features and Euclidean k-means with a maximum of 100 iterations. The representative grid is $K \in \{1, 5, 25\}$ and the neighbor grid is $\{1, 3, 5\}$. FeNeC-M estimates one covariance per class, applies the author's ViT single-shrinkage and correlation normalization with $(\gamma_1, \gamma_2) \in \{(1, 1), (6.12, 8.10), (9.98, 0)\}$, then uses inverse squared-Mahalanobis-distance votes. FeNeC-E uses inverse unsquared-Euclidean-distance votes. Their native predictions are preserved when votes are converted to log votes, with a floor of $10^{-30}$. The calibrated forms select a novel offset from $[-4, 4]$ in steps of 0.25 on the pseudo episode. Harmonic macro accuracy chooses the configuration.

The five-shot protocol caps a class's representative count at its sample count. A class with one fitting sample uses identity covariance, avoiding an undefined sample covariance. These are explicit sparse-data adaptations. A per-class independent summary can be cached before its arrival; the voting neighborhood always contains only currently admitted classes. Native and calibrated forms share each current-vocabulary score matrix, so their complete matched-retention frontiers coincide, while fixed-offset trajectories can differ. The final grids are documented with the code. FeNeC frontiers use its sparse global-neighbor votes, with unvoted classes at the common floor; feasibility refers to this output interface.

All ImageNet-R images are content-hashed before pool assignment. Fifty-nine byte-identical duplicates are removed globally. No retained content hash appears in two fitting, validation, support or test pools. The source archive has SHA-256 `18c6bf493b39a0d975d48e587437f562caab9c52ae6327dcfa9dd8eb54aa1b52`. The class-name mapping, image paths, achieved fitting counts, source hashes, calibrated settings and prediction margins accompany the code and records. These are task-specific splits of ImageNet-R, whose source release is a rendition benchmark [20].

Table S8: ImageNet-R matched-retention recognition at three budgets after five arrivals. Values are percentages, mean ± partition-level sample standard deviation over feasible partitions. Counts specify available partition sets.

| Budget | Readout | Base | Novel | Feasible |
|---|---|---|---|---|
| 90% | Retained classifier | 81.83 ± 0.29 | 86.80 ± 0.27 | 3/3 |
| 90% | Classifier removed (fixed gates) | 81.81 ± 0.31 | 85.53 ± 0.49 | 3/3 |
| 90% | No classifier (recalibrated) | 81.81 ± 0.31 | 85.34 ± 0.46 | 3/3 |
| 90% | Anchored text + prototypes | 81.83 ± 0.32 | 86.59 ± 0.58 | 3/3 |
| 90% | Tip-Adapter sum + offset | 81.82 ± 0.30 | 83.74 ± 0.25 | 3/3 |
| 90% | Tip-Adapter mean + offset | 81.83 ± 0.30 | 85.30 ± 0.23 | 3/3 |
| 95% | Retained classifier | 86.38 ± 0.32 | 84.48 ± 0.72 | 3/3 |
| 95% | Classifier removed (fixed gates) | 86.37 ± 0.31 | 72.62 ± 12.51 | 3/3 |
| 95% | No classifier (recalibrated) | 86.37 ± 0.31 | 72.69 ± 12.57 | 3/3 |
| 95% | Anchored text + prototypes | 86.37 ± 0.32 | 83.32 ± 1.40 | 3/3 |
| 95% | Tip-Adapter sum + offset | 86.41 ± 0.30 | 70.93 ± 10.32 | 3/3 |
| 95% | Tip-Adapter mean + offset | 86.38 ± 0.33 | 81.58 ± 1.72 | 3/3 |
| 99% | Retained classifier | 90.02 ± 0.34 | 73.69 ± 1.50 | 3/3 |
| 99% | Classifier removed (fixed gates) | – | – | 0/3 |
| 99% | No classifier (recalibrated) | – | – | 0/3 |
| 99% | Anchored text + prototypes | 89.98 ± 0.47 | 47.62 ± 13.36 | 2/3 |
| 99% | Tip-Adapter sum + offset | – | – | 0/3 |
| 99% | Tip-Adapter mean + offset | – | – | 0/3 |

Table S9: ImageNet-R test outcomes with the 95% base-validation-only threshold policy. Validation feasibility is reported separately from achieved test accuracy. Tip-Adapter uses its base-only selected calibrated cache settings, followed by validation-only threshold selection.

| Readout | Base test | Novel test | Validation feasible |
|---|---|---|---|
| Retained classifier | 86.01 ± 0.59 | 84.73 ± 0.71 | 3/3 |
| Classifier removed (fixed gates) | 86.38 ± 0.18 | 57.35 ± 38.51 | 3/3 |
| No classifier (recalibrated) | 86.48 ± 0.20 | 57.10 ± 38.30 | 3/3 |
| Anchored text + prototypes | 86.36 ± 0.47 | 83.26 ± 1.49 | 3/3 |
| Tip-Adapter sum + offset | 86.10 ± 0.46 | 74.08 ± 5.48 | 3/3 |
| Tip-Adapter mean + offset | 86.68 ± 0.23 | 80.75 ± 2.09 | 3/3 |

Table S10: FeNeC matched-test-retention operating points at strict budgets. Values are percentages, mean ± sample standard deviation over feasible partitions; missing values have zero feasible partitions. Native and offset-calibrated forms share a score matrix and hence the same complete offset-sweep frontier.

| Data | Budget | Readout | Base | Novel | Feasible |
|---|---|---|---|---|---|
| CIFAR-100 | 95% | FeNeC-M + offset | – | – | 0/3 |
| CIFAR-100 | 95% | FeNeC-E + offset | – | – | 0/3 |
| CIFAR-100 | 99% | FeNeC-M + offset | – | – | 0/3 |
| CIFAR-100 | 99% | FeNeC-E + offset | – | – | 0/3 |
| Pets | 95% | FeNeC-M + offset | – | – | 0/3 |
| Pets | 95% | FeNeC-E + offset | 94.23 | 69.46 | 1/3 |
| Pets | 99% | FeNeC-M + offset | – | – | 0/3 |
| Pets | 99% | FeNeC-E + offset | – | – | 0/3 |
| ImageNet-R | 95% | FeNeC-M + offset | – | – | 0/3 |
| ImageNet-R | 95% | FeNeC-E + offset | – | – | 0/3 |
| ImageNet-R | 99% | FeNeC-M + offset | – | – | 0/3 |
| ImageNet-R | 99% | FeNeC-E + offset | – | – | 0/3 |

## Appendix G. Dense caches, repeated calibration and measured readout cost

*Appendix G.1. Dense semantic-cache protocol*

The training-free Tip-Adapter score follows the ECCV paper and the author implementation [17]: cosine similarity to labeled cache keys is transformed by $\exp[-\beta(1 - \text{similarity})]$, summed into classes and added to 100 times text similarity with weight $\alpha$. The author code is pinned at commit `d0e2d6f8c5feb8b6ce937b757810761f7155d4d5`. All permitted base fitting images and five reserved novel support images supply the keys. The sum variant retains the published aggregation. A separate class-mean variant divides each class's retrieval score by its key count, isolating the effect of unequal long-tailed cache sizes. Both reuse the same single-view frozen features and text anchors as the tables. This comparison transfers the published readout under controlled feature access; the original augmented-cache extraction and trainable Tip-Adapter-F are distinct settings.

On each original base-only pseudo episode, we search $\beta \in \{0.1, 1, 2, 5, 10, 20, 50\}$ and $\alpha \in \{0.1, 1, 5, 10, 20, 50, 100, 200\}$. The native form selects both parameters with zero offset. The calibrated form jointly selects these parameters and an offset from $[-40, 40]$ in steps of 2.5, using macro harmonic accuracy. This 33-point offset grid reflects the raw 100-scaled CLIP score system. No actual novel query or test label is used in selection. Both forms are reported because their selected cache settings can differ. All matched-retention targets are the original retained readout's partition-specific absolute targets.

Table S11: Complete five-shot fixed-gate readout comparison. Values are percentages, mean ± sample standard deviation over three class partitions after averaging the two arrival orders. The paired gain is retained minus comparator average incremental accuracy; a negative gain favors the comparator. Native and calibrated cache forms are selected separately.

| Data | Readout | Incremental avg. | Final | Paired avg. gain |
|---|---|---|---|---|
| CIFAR-100 | Retained classifier | 82.02 ± 0.84 | 79.31 ± 1.14 | 0.00 ± 0.00 |
| CIFAR-100 | Classifier removed (fixed gates) | 71.40 ± 14.78 | 68.03 ± 15.63 | 10.62 ± 15.22 |
| CIFAR-100 | No classifier (recalibrated) | 79.84 ± 0.82 | 77.56 ± 1.02 | 2.18 ± 0.24 |
| CIFAR-100 | Anchored text + prototypes | 81.56 ± 0.52 | 79.30 ± 0.59 | 0.46 ± 0.44 |
| CIFAR-100 | Tip-Adapter sum + offset | 79.57 ± 0.41 | 77.21 ± 0.38 | 2.45 ± 0.63 |
| CIFAR-100 | Tip-Adapter mean + offset | 81.74 ± 0.93 | 79.40 ± 1.11 | 0.28 ± 1.07 |
| CIFAR-100 | FeNeC-M + offset | 66.60 ± 1.34 | 63.65 ± 1.55 | 15.42 ± 0.49 |
| CIFAR-100 | FeNeC-E + offset | 72.63 ± 1.34 | 69.00 ± 1.97 | 9.39 ± 0.68 |
| CIFAR-100 | FeNeC-M (native) | 59.11 ± 0.85 | 53.58 ± 1.27 | 22.91 ± 0.31 |
| CIFAR-100 | FeNeC-E (native) | 72.63 ± 1.34 | 69.00 ± 1.97 | 9.39 ± 0.68 |
| CIFAR-100 | Tip-Adapter sum (native) | 79.57 ± 0.41 | 77.21 ± 0.38 | 2.45 ± 0.63 |
| CIFAR-100 | Tip-Adapter mean (native) | 81.74 ± 0.93 | 79.40 ± 1.11 | 0.28 ± 1.07 |
| Pets | Retained classifier | 94.09 ± 2.37 | 91.77 ± 3.16 | 0.00 ± 0.00 |
| Pets | Classifier removed (fixed gates) | 77.17 ± 26.01 | 76.85 ± 23.25 | 16.92 ± 28.33 |
| Pets | No classifier (recalibrated) | 94.20 ± 2.05 | 92.57 ± 2.31 | -0.11 ± 0.56 |
| Pets | Anchored text + prototypes | 94.35 ± 0.60 | 92.48 ± 1.10 | -0.26 ± 1.82 |
| Pets | Tip-Adapter sum + offset | 93.38 ± 2.32 | 91.50 ± 2.78 | 0.71 ± 1.21 |
| Pets | Tip-Adapter mean + offset | 94.47 ± 2.38 | 92.88 ± 2.51 | -0.37 ± 1.61 |
| Pets | FeNeC-M + offset | 80.34 ± 1.29 | 75.66 ± 1.72 | 13.76 ± 1.61 |
| Pets | FeNeC-E + offset | 84.43 ± 2.37 | 79.82 ± 2.35 | 9.67 ± 1.50 |
| Pets | FeNeC-M (native) | 73.66 ± 4.73 | 66.74 ± 6.13 | 20.44 ± 2.69 |
| Pets | FeNeC-E (native) | 84.43 ± 2.37 | 79.82 ± 2.35 | 9.67 ± 1.50 |
| Pets | Tip-Adapter sum (native) | 94.35 ± 1.08 | 92.72 ± 1.65 | -0.25 ± 1.32 |
| Pets | Tip-Adapter mean (native) | 95.15 ± 1.28 | 93.74 ± 1.29 | -1.06 ± 1.33 |
| ImageNet-R | Retained classifier | 87.23 ± 0.21 | 85.40 ± 0.36 | 0.00 ± 0.00 |
| ImageNet-R | Classifier removed (fixed gates) | 77.13 ± 6.75 | 74.64 ± 7.19 | 10.09 ± 6.55 |
| ImageNet-R | No classifier (recalibrated) | 85.01 ± 0.57 | 83.65 ± 0.70 | 2.22 ± 0.40 |
| ImageNet-R | Anchored text + prototypes | 86.36 ± 0.24 | 84.67 ± 0.39 | 0.87 ± 0.29 |
| ImageNet-R | Tip-Adapter sum + offset | 84.62 ± 0.08 | 82.94 ± 0.07 | 2.61 ± 0.21 |
| ImageNet-R | Tip-Adapter mean + offset | 85.75 ± 0.30 | 84.19 ± 0.35 | 1.48 ± 0.35 |
| ImageNet-R | FeNeC-M + offset | 59.12 ± 0.54 | 55.79 ± 0.63 | 28.11 ± 0.46 |
| ImageNet-R | FeNeC-E + offset | 69.66 ± 0.97 | 66.09 ± 0.62 | 17.56 ± 0.80 |
| ImageNet-R | FeNeC-M (native) | 54.57 ± 0.47 | 49.71 ± 0.32 | 32.65 ± 0.37 |
| ImageNet-R | FeNeC-E (native) | 69.66 ± 0.97 | 66.09 ± 0.62 | 17.56 ± 0.80 |
| ImageNet-R | Tip-Adapter sum (native) | 84.62 ± 0.08 | 82.94 ± 0.07 | 2.61 ± 0.21 |
| ImageNet-R | Tip-Adapter mean (native) | 85.75 ± 0.30 | 84.19 ± 0.35 | 1.48 ± 0.35 |

*Appendix G.2. Repeated calibration and support budgets*

The 81 configurations span three datasets, three original class partitions, $k \in \{1, 2, 5\}$ and three joint calibration/support trials. Trial zero reuses the original pseudo split. The other trials select $\lfloor |\mathcal{B}|/2 \rfloor$ pseudo-base classes with seeds $1701 + p$ and $2701 + p$. Pseudo classifiers select ridge regularization on pseudo-base validation. Support-index seeds $7, 8, 9$ select one- and two-shot subsets from each five-image reserve. At five shots, the support set is identical across trials: variation comes from the pseudo split. These are correlated trials of the same underlying class partitions. We average the orders and trials within each partition and report the mean and sample standard deviation of the three partition means.

Each configuration independently calibrates the retained, recalibrated-removal, dense-prototype and dynamic-reference readouts on the same base-only episode. Fixed-gate removal preserves the retained mixture's remaining parameters. Dynamic normalization uses the current text candidates and current novel prototypes as references, with the same learned branch and calibration grid as the retained table. Its own mixture is selected independently and stays fixed across arrivals. All retention targets are shared within each configuration. Feasibility counts refer to nine partition/trial combinations per support budget. When only some trials are feasible, the group accuracies average those trials within each partition before summarizing partitions; paired constrained comparisons use the intersection of feasible trial IDs.

Fixed references have zero restricted pre-offset candidate-ranking changes in all 648 append checks (81 configurations, two orders, four appends). The largest floating-point score differences are $1.40 \times 10^{-4}$ on CIFAR, $5.36 \times 10^{-6}$ on

Table S12: Calibration and support robustness. Average incremental and final accuracies use fixed gates. Paired gains are retained minus each independently calibrated comparator, after averaging two arrival orders and three trials within each partition. Values are percentages or percentage points, mean ± sample standard deviation across three partitions.

| Data | Shots | Retained avg. | Δ recalibrated | Δ dense | Δ dynamic | Retained final |
|---|---|---|---|---|---|---|
| CIFAR-100 | 1 | 79.77 ± 0.61 | 0.55 ± 1.15 | 0.08 ± 0.72 | 0.10 ± 0.74 | 76.32 ± 1.22 |
| CIFAR-100 | 2 | 80.20 ± 0.32 | 0.84 ± 0.67 | -0.76 ± 0.52 | 0.55 ± 0.84 | 77.00 ± 0.85 |
| CIFAR-100 | 5 | 80.57 ± 0.96 | 1.22 ± 1.18 | -0.79 ± 1.08 | 0.62 ± 0.59 | 77.25 ± 1.09 |
| Pets | 1 | 93.94 ± 2.37 | 0.05 ± 0.67 | -0.24 ± 0.48 | -0.54 ± 0.11 | 91.60 ± 3.20 |
| Pets | 2 | 92.64 ± 1.82 | -1.45 ± 1.19 | -1.62 ± 1.15 | -1.43 ± 1.09 | 90.09 ± 1.87 |
| Pets | 5 | 94.56 ± 1.74 | 0.78 ± 1.03 | 0.43 ± 0.80 | 0.47 ± 1.43 | 92.75 ± 2.00 |
| ImageNet-R | 1 | 86.16 ± 0.22 | 1.81 ± 0.33 | 1.01 ± 0.29 | -0.07 ± 0.29 | 83.83 ± 0.14 |
| ImageNet-R | 2 | 86.58 ± 0.37 | 2.10 ± 0.28 | 0.62 ± 0.34 | -0.07 ± 0.28 | 84.43 ± 0.67 |
| ImageNet-R | 5 | 87.16 ± 0.12 | 2.14 ± 0.37 | 0.54 ± 0.23 | 0.15 ± 0.19 | 85.36 ± 0.25 |

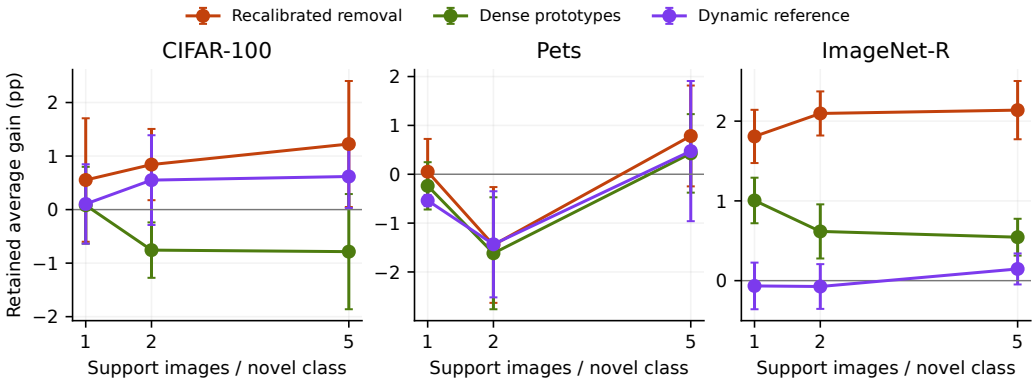



Figure S4: Paired average incremental gains under one-, two- and five-shot support. Error bars are partition-level sample standard deviations after averaging correlated calibration/support trials and arrival orders. The horizontal zero line identifies which readout has higher average accuracy in each setting.

Pets and $2.19 \times 10^{-5}$ on ImageNet-R. Dynamic references produce maximum score changes of 2.40, 1.78 and 1.30, respectively, and up to 429, 206 and 118 restricted prediction changes per append over the full cached test query pool. These diagnostics compare the same previously active candidate IDs before adding the fixed group offset; they isolate changes induced by reference statistics. Recognition accuracy is reported separately in the fixed-gate and retention-policy tables.

Table S13: CIFAR-100: final-session outcomes in the repeated-calibration study at the 95% budget. Match uses base test labels descriptively; Val. uses base validation only and reports achieved test accuracy. Feasibility is counted over nine correlated partition/trial combinations.

| Shots | Readout | Match base | Match novel | Feasible | Val. base | Val. novel | Feasible |
|---|---|---|---|---|---|---|---|
| 1 | Retained classifier | 83.54 ± 0.81 | 73.03 ± 3.08 | 9/9 | 80.77 ± 2.34 | 76.30 ± 3.82 | 9/9 |
| 1 | No classifier (recalibrated) | 83.37 ± 0.57 | 59.17 ± 3.51 | 8/9 | 80.92 ± 1.45 | 72.28 ± 4.17 | 9/9 |
| 1 | Anchored text + prototypes | 83.37 ± 0.57 | 63.65 ± 3.03 | 8/9 | 80.38 ± 2.28 | 73.74 ± 5.16 | 9/9 |
| 1 | Dynamic reference | 83.54 ± 0.81 | 73.07 ± 3.81 | 9/9 | 81.10 ± 2.54 | 75.86 ± 4.56 | 9/9 |
| 2 | Retained classifier | 82.99 ± 0.46 | 73.84 ± 3.52 | 9/9 | 80.65 ± 1.93 | 76.40 ± 4.33 | 9/9 |
| 2 | No classifier (recalibrated) | 82.99 ± 0.46 | 61.20 ± 0.62 | 9/9 | 81.14 ± 1.01 | 71.44 ± 2.35 | 9/9 |
| 2 | Anchored text + prototypes | 82.99 ± 0.46 | 71.62 ± 6.74 | 9/9 | 80.81 ± 1.65 | 76.62 ± 4.51 | 9/9 |
| 2 | Dynamic reference | 82.99 ± 0.46 | 74.70 ± 3.56 | 9/9 | 80.37 ± 1.53 | 77.66 ± 2.64 | 9/9 |
| 5 | Retained classifier | 83.83 ± 0.76 | 75.71 ± 2.77 | 9/9 | 82.03 ± 1.38 | 77.69 ± 1.64 | 9/9 |
| 5 | No classifier (recalibrated) | 83.55 ± 0.29 | 62.19 ± 6.29 | 7/9 | 81.21 ± 1.18 | 73.83 ± 6.90 | 7/9 |
| 5 | Anchored text + prototypes | 83.83 ± 0.76 | 73.28 ± 3.14 | 9/9 | 81.95 ± 0.54 | 75.45 ± 1.14 | 8/9 |
| 5 | Dynamic reference | 83.83 ± 0.76 | 73.83 ± 0.74 | 9/9 | 82.44 ± 1.94 | 76.58 ± 2.18 | 9/9 |

Table S14: Oxford-IIIT Pets: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

| Shots | Readout | Match base | Match novel | Feasible | Val. base | Val. novel | Feasible |
|---|---|---|---|---|---|---|---|
| 1 | Retained classifier | 92.58 ± 1.83 | 93.50 ± 0.89 | 9/9 | 88.07 ± 1.16 | 94.84 ± 0.89 | 9/9 |
| 1 | No classifier (recalibrated) | 92.58 ± 1.85 | 92.30 ± 2.36 | 9/9 | 88.56 ± 1.16 | 95.45 ± 0.70 | 6/9 |
| 1 | Anchored text + prototypes | 92.57 ± 1.84 | 93.03 ± 1.60 | 9/9 | 87.59 ± 1.34 | 94.59 ± 1.55 | 9/9 |
| 1 | Dynamic reference | 92.56 ± 1.82 | 93.74 ± 1.02 | 9/9 | 87.89 ± 1.50 | 95.47 ± 0.16 | 8/9 |
| 2 | Retained classifier | 92.28 ± 2.51 | 94.42 ± 1.14 | 9/9 | 88.05 ± 2.52 | 95.66 ± 0.49 | 9/9 |
| 2 | No classifier (recalibrated) | 92.27 ± 2.52 | 93.97 ± 0.96 | 9/9 | 87.04 ± 3.11 | 95.89 ± 0.27 | 7/9 |
| 2 | Anchored text + prototypes | 92.27 ± 2.50 | 94.33 ± 0.58 | 9/9 | 87.64 ± 2.55 | 95.51 ± 0.58 | 9/9 |
| 2 | Dynamic reference | 92.27 ± 2.50 | 94.62 ± 1.31 | 9/9 | 87.85 ± 2.40 | 95.77 ± 0.27 | 9/9 |
| 5 | Retained classifier | 92.40 ± 2.41 | 94.95 ± 1.39 | 9/9 | 88.31 ± 2.40 | 96.06 ± 0.47 | 9/9 |
| 5 | No classifier (recalibrated) | 92.40 ± 2.42 | 94.08 ± 0.92 | 9/9 | 87.39 ± 2.71 | 96.12 ± 0.41 | 7/9 |
| 5 | Anchored text + prototypes | 92.39 ± 2.39 | 94.79 ± 0.92 | 9/9 | 87.94 ± 2.68 | 95.99 ± 0.80 | 9/9 |
| 5 | Dynamic reference | 92.40 ± 2.40 | 94.84 ± 1.36 | 9/9 | 87.44 ± 2.34 | 96.12 ± 0.27 | 9/9 |

Table S15: ImageNet-R: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

| Shots | Readout | Match base | Match novel | Feasible | Val. base | Val. novel | Feasible |
|---|---|---|---|---|---|---|---|
| 1 | Retained classifier | 86.24 ± 0.38 | 81.59 ± 0.50 | 9/9 | 85.73 ± 0.71 | 81.95 ± 0.67 | 9/9 |
| 1 | No classifier (recalibrated) | 86.25 ± 0.37 | 70.69 ± 10.24 | 9/9 | 86.13 ± 0.37 | 69.56 ± 13.12 | 9/9 |
| 1 | Anchored text + prototypes | 86.24 ± 0.36 | 78.57 ± 0.59 | 9/9 | 86.14 ± 0.54 | 78.97 ± 0.47 | 9/9 |
| 1 | Dynamic reference | 86.25 ± 0.36 | 81.78 ± 0.76 | 9/9 | 85.87 ± 0.61 | 82.11 ± 1.07 | 9/9 |
| 2 | Retained classifier | 86.38 ± 0.34 | 82.67 ± 1.21 | 9/9 | 86.10 ± 0.63 | 82.86 ± 1.30 | 9/9 |
| 2 | No classifier (recalibrated) | 86.38 ± 0.33 | 69.12 ± 14.06 | 9/9 | 86.27 ± 0.38 | 58.73 ± 33.47 | 9/9 |
| 2 | Anchored text + prototypes | 86.39 ± 0.34 | 81.53 ± 2.15 | 9/9 | 86.40 ± 0.40 | 81.50 ± 1.68 | 9/9 |
| 2 | Dynamic reference | 86.38 ± 0.33 | 82.80 ± 0.65 | 9/9 | 86.04 ± 0.67 | 83.02 ± 0.76 | 9/9 |
| 5 | Retained classifier | 86.35 ± 0.27 | 84.48 ± 0.35 | 9/9 | 86.08 ± 0.44 | 84.69 ± 0.46 | 9/9 |
| 5 | No classifier (recalibrated) | 86.34 ± 0.27 | 73.89 ± 9.72 | 9/9 | 86.36 ± 0.21 | 63.98 ± 26.37 | 9/9 |
| 5 | Anchored text + prototypes | 86.34 ± 0.27 | 83.45 ± 1.62 | 9/9 | 86.23 ± 0.52 | 83.65 ± 1.19 | 9/9 |
| 5 | Dynamic reference | 86.35 ± 0.29 | 84.42 ± 0.56 | 9/9 | 86.21 ± 0.47 | 84.50 ± 0.69 | 9/9 |

### *Appendix G.3. Readout cost and restart verification*

Table S16 measures 512 cached queries on the RTX 4070 Laptop in float32. Each readout receives five warm-ups and 20 synchronized timing repetitions. Static cache values are constructed before timing, query chunks contain 256 images for retrieval, and unused score branches are skipped. An independently checked implementation verifies score agreement within $10^{-3}$ for every optimized readout. Persistent tensor bytes count text anchors, learned weights and references/prototypes for the tables, or text anchors, cache keys and the stored dense one-hot/class-mean value matrix for retrieval. These are implementation-level readout costs, excluding the common image encoder, query features and temporary workspaces. They do not represent complete model memory or end-to-end image inference time.

Every calibration and arrival-session result is written atomically. A live pause/resume check preserved a completed session's SHA-256 hash and modification time, then completed all 90 configuration units. The numerical verifier reconstructs group predictions, macro weights, selected thresholds and test metrics from archived sufficient statistics, including the base-validation

Table S16: Measured readout latency and persistent tensor storage for a final-session 512-query batch on the same laptop GPU. MiB denotes $2^{20}$ bytes. Cache keys number 4,937 on CIFAR, 553 on Pets and 3,830 on ImageNet-R in partition zero.

| Data | Readout | Median (ms) | P90 (ms) | Stored tensors (MiB) |
|---|---|---|---|---|
| CIFAR-100 | Retained classifier | 0.715 | 0.763 | 0.586 |
| CIFAR-100 | Dense text + prototypes | 0.515 | 0.542 | 0.469 |
| CIFAR-100 | Tip-Adapter sum | 0.838 | 0.862 | 16.205 |
| CIFAR-100 | Tip-Adapter mean | 0.841 | 0.857 | 16.205 |
| Pets | Retained classifier | 0.880 | 1.280 | 0.220 |
| Pets | Dense text + prototypes | 0.586 | 0.680 | 0.176 |
| Pets | Tip-Adapter sum | 0.564 | 0.673 | 1.771 |
| Pets | Tip-Adapter mean | 0.569 | 0.732 | 1.771 |
| ImageNet-R | Retained classifier | 0.864 | 0.918 | 1.465 |
| ImageNet-R | Dense text + prototypes | 0.631 | 0.790 | 1.172 |
| ImageNet-R | Tip-Adapter sum | 0.759 | 0.873 | 14.729 |
| ImageNet-R | Tip-Adapter mean | 0.732 | 0.898 | 14.729 |

selection arrays. The trial-zero five-shot table results agree exactly with the prior three-dataset records. Source, protocol and cache hashes, all outcomes and reproduction commands accompany the release.

## Appendix H. Validation-query budget for admission thresholds

This experiment varies the old-class query labels used to select the admission threshold, conditional on the original five-shot classifier and fusion calibration. All fitted parameters, cache settings and branch mixtures stay fixed. The evaluation uses the original trial-zero score archives, class partitions and arrival order zero on each dataset. No image encoder or new classifier is fitted.

For each base class with $n_c$ validation images, the nominal fraction $f$ keeps $\max(1, \lfloor f n_c \rfloor)$ images. Fractions $f \in \{0.25, 0.5\}$ use three predeclared subset seeds, with nested class-wise permutations; the full budget uses all images once. Actual query counts are reported because rounding and the one-query

minimum change the realized fractions. Macro weights are recomputed from the selected class counts. The retained readout's closed-base accuracy on that subset defines a shared target $rA^{\text{closed}}_{B,\text{ref,subset}}$ for all six readouts, with $r \in \{0.95, 0.99\}$. The largest feasible offset uses only the selected base-validation margins and labels.

After selection, test queries measure both group accuracies. A test target is met when $A_{B,\text{test}} \geq rA^{\text{closed}}_{B,\text{ref,test}}$; this criterion is evaluated after the offset is fixed and never enters threshold selection. Table S17 separates validation feasibility from test target attainment. Subset repeats are averaged within each class partition before the mean and sample standard deviation over available partition means. The nine reduced-budget cases share three underlying partitions and are not independent repetitions. Small per-class counts quantize the selected threshold, so different target ratios can yield identical predictions and increasing the validation fraction need not change test target attainment monotonically. These empirical policies do not supply a finite-sample guarantee on test retention.

The full-budget policies reproduce the original validation-selected outcomes exactly. Saved predictions reconstruct every offset, macro accuracy and attainment flag across 315 session units and 3,780 metric records. The complete budgets, query fractions and test-retention ratios are in `validation_budget_summary.csv`. This experiment varies threshold-selection queries while holding the initial full-data fusion calibration fixed.

Table S17: Retained readout under different validation-query budgets after the final arrival. Accuracies are percentages, mean ± partition-level sample standard deviation after averaging subset repeats. Queries gives the actual old-class validation count range. Val. feasible counts partition/subset combinations meeting the validation constraint; Test met counts feasible combinations meeting the separate test criterion. The full budget uses one subset per partition.

| Data | Nominal budget | Queries | Target | Base test | Novel test | Val. feasible | Test met |
|---|---|---|---|---|---|---|---|
| CIFAR-100 | 25% | 40 | 95% | 82.46 ± 2.25 | 75.14 ± 3.50 | 9/9 | 4/9 |
| CIFAR-100 | 25% | 40 | 99% | 85.21 ± 2.64 | 66.99 ± 9.18 | 9/9 | 4/9 |
| CIFAR-100 | 50% | 68 | 95% | 81.94 ± 3.23 | 76.92 ± 3.72 | 9/9 | 3/9 |
| CIFAR-100 | 50% | 68 | 99% | 85.85 ± 2.61 | 64.30 ± 10.71 | 9/9 | 5/9 |
| CIFAR-100 | 100% | 161–163 | 95% | 82.99 ± 2.29 | 75.71 ± 3.14 | 3/3 | 1/3 |
| CIFAR-100 | 100% | 161–163 | 99% | 86.48 ± 1.65 | 62.83 ± 10.11 | 3/3 | 2/3 |
| Pets | 25% | 15 | 95% | 91.77 ± 2.39 | 92.42 ± 2.64 | 9/9 | 5/9 |
| Pets | 25% | 15 | 99% | 91.77 ± 2.39 | 92.42 ± 2.64 | 9/9 | 4/9 |
| Pets | 50% | 25 | 95% | 92.02 ± 0.95 | 94.75 ± 1.23 | 9/9 | 5/9 |
| Pets | 50% | 25 | 99% | 95.05 ± 0.48 | 90.49 ± 5.09 | 9/9 | 5/9 |
| Pets | 100% | 60 | 95% | 90.29 ± 2.90 | 95.68 ± 0.83 | 3/3 | 0/3 |
| Pets | 100% | 60 | 99% | 96.43 ± 2.15 | 88.67 ± 7.89 | 3/3 | 2/3 |
| ImageNet-R | 25% | 696–731 | 95% | 85.85 ± 0.38 | 84.98 ± 0.38 | 9/9 | 2/9 |
| ImageNet-R | 25% | 696–731 | 99% | 89.79 ± 0.16 | 74.81 ± 2.33 | 9/9 | 5/9 |
| ImageNet-R | 50% | 1444–1519 | 95% | 85.93 ± 0.65 | 84.87 ± 0.63 | 9/9 | 1/9 |
| ImageNet-R | 50% | 1444–1519 | 99% | 89.90 ± 0.33 | 74.24 ± 2.68 | 9/9 | 5/9 |
| ImageNet-R | 100% | 2938–3087 | 95% | 86.01 ± 0.59 | 84.73 ± 0.71 | 3/3 | 0/3 |
| ImageNet-R | 100% | 2938–3087 | 99% | 89.94 ± 0.25 | 74.48 ± 1.35 | 3/3 | 2/3 |

Table S18: All readouts at the nominal 25% validation-query budget and 95% target after the final arrival. All classifiers, mixtures and cache settings are held fixed from the original five-shot calibration. Accuracy aggregation and feasibility/attainment counts follow Table S17. Missing metrics correspond to zero feasible subsets.

| Data | Readout | Base test | Novel test | Val. feasible | Test met |
|---|---|---|---|---|---|
| CIFAR-100 | Retained classifier | 82.46 ± 2.25 | 75.14 ± 3.50 | 9/9 | 4/9 |
| CIFAR-100 | Fixed-gate removal | 79.39 ± 2.08 | 74.84 ± 0.15 | 5/9 | 1/5 |
| CIFAR-100 | Recalibrated removal | 79.48 ± 2.19 | 73.22 ± 2.45 | 5/9 | 1/5 |
| CIFAR-100 | Dense text + prototypes | 81.30 ± 2.08 | 77.12 ± 2.64 | 7/9 | 0/7 |
| CIFAR-100 | Tip-Adapter sum | 79.32 ± 2.30 | 71.40 ± 7.86 | 6/9 | 1/6 |
| CIFAR-100 | Tip-Adapter mean | 79.51 ± 3.58 | 77.12 ± 5.00 | 8/9 | 1/8 |
| Pets | Retained classifier | 91.77 ± 2.39 | 92.42 ± 2.64 | 9/9 | 5/9 |
| Pets | Fixed-gate removal | 88.87 ± 3.95 | 92.77 ± 5.91 | 6/9 | 1/6 |
| Pets | Recalibrated removal | 87.68 ± 3.95 | 89.64 ± 11.35 | 6/9 | 1/6 |
| Pets | Dense text + prototypes | 88.50 ± 6.27 | 93.41 ± 3.82 | 7/9 | 2/7 |
| Pets | Tip-Adapter sum | 89.86 ± 3.58 | 89.59 ± 5.68 | 7/9 | 2/7 |
| Pets | Tip-Adapter mean | 90.25 ± 1.80 | 91.90 ± 3.31 | 7/9 | 2/7 |
| ImageNet-R | Retained classifier | 85.85 ± 0.38 | 84.98 ± 0.38 | 9/9 | 2/9 |
| ImageNet-R | Fixed-gate removal | 85.66 ± 0.86 | 79.23 ± 1.90 | 7/9 | 3/7 |
| ImageNet-R | Recalibrated removal | 85.74 ± 0.90 | 79.08 ± 1.74 | 7/9 | 3/7 |
| ImageNet-R | Dense text + prototypes | 86.39 ± 0.50 | 83.05 ± 0.22 | 9/9 | 4/9 |
| ImageNet-R | Tip-Adapter sum | 85.47 ± 0.89 | 77.71 ± 1.44 | 8/9 | 4/8 |
| ImageNet-R | Tip-Adapter mean | 86.49 ± 0.54 | 81.22 ± 0.63 | 9/9 | 6/9 |

## Appendix I. Ten-partition replication, encoder transfer and simple fusion

The expanded experiment fixes class-partition seeds $0, \ldots, 9$ on each of the three existing image pools. Assignments and pseudo splits $0, 1, 2$ are retained exactly, and the original L/14 results reconstruct with maximum metric difference zero. New seeds use the same frequency-stratified allocation, base/novel counts, five-shot reserve, pseudo-episode selection and two arrival-order rules. These are ten distinct class assignments on shared images, not ten independent datasets. Primary fixed-gate comparisons average both orders within a partition. The conditional paired percentile bootstrap uses 20,000 resamples of the ten paired partition values with seed 2026. Constrained differences use identical jointly feasible partition/order/session cells. JSON records retain the partition values and available sets, and CSV tables report every budget.

*Appendix I.1. Second encoder and access controls*

OpenAI CLIP ViT-B/32 uses cached revision `3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268` with 512-dimensional projected features; L/14 uses 768 dimensions. Both consume the exact same fitting, validation, support and test images, dataset-specific preprocessing and eight text templates. Text anchors and ridge columns are recomputed for the selected encoder; no task-adapted features are used. The two encoders have separate closed-base references and hence separate absolute targets. Config, weights, tokenizer and pool hashes are recorded. Batch-16 feature shards support atomic restart with contiguous-index guards, and eight fresh test images are re-encoded per dataset to check each completed B/32 cache.

*Appendix I.2. Raw and fixed-temperature controls*

Raw fusion uses $q_T$, $q_L$ and $q_P$ directly in the same availability pattern as the retained table. Fixed-temperature fusion uses $q_e/\sigma_e$, where

$$\sigma_e = \max\left(10^{-6}, \frac{1}{|\mathcal{V}_B|}\sum_{x\in\mathcal{V}_B} \mathrm{Std}_{r\in\mathcal{R}_e}\, q_e(x,r)\right).$$

For the pseudo episode, $\mathcal{V}_B$ contains base-validation queries and $\mathcal{R}_e$ contains pseudo-base reference columns. Full-model scales use full base columns and base validation before arrivals. The three scalars remain fixed across sessions. Coefficient and offset grids are identical to the main retained model, with the same harmonic pseudo-episode objective. Neither control requires current-vocabulary statistics; both can preserve old scores. Their performance tests the empirical value of the per-query normalization, conditional on this specified calibration grid.

*Appendix I.3. Full matched and validation-policy results*

The main validation table includes retained, recalibrated-removal, dense-prototype, class-mean cache and simple-fusion alternatives. Summed cache and fixed-gate removal remain in the complete records. The following tables give both encoders' matched-test frontiers and the B/32 validation policy at the 95% budget. Every feasible set and test-attainment count refers to the final active vocabulary; order zero counts each partition once. Because means may cover different feasible sets, an unpaired difference of available means is not reported as a paired gain. Complete 80/85/90/95/99% outcomes are retained in `revision_summary.csv`; per-partition raw predictions supply both test and validation sufficient statistics.

Table S19: Complete eight-readout ViT-L/14 validation-selected 95% policy. Accuracies are achieved test percentages, mean ± sample standard deviation over feasible partitions. Feasible and Test met have the same meanings as the main validation-policy table.

| Data | Readout | Base | Novel | Feasible | Test met |
|---|---|---|---|---|---|
| CIFAR-100 | Retained references | 84.26 ± 1.91 | 77.10 ± 2.53 | 10/10 | 7/10 |
| CIFAR-100 | Fixed-gate removal | 83.05 ± 1.64 | 65.32 ± 12.41 | 6/10 | 1/6 |
| CIFAR-100 | Recalibrated removal | 82.91 ± 1.62 | 66.03 ± 13.04 | 6/10 | 1/6 |
| CIFAR-100 | Dense text + prototypes | 84.00 ± 1.76 | 75.10 ± 7.15 | 8/10 | 4/8 |
| CIFAR-100 | Raw-score fusion | 82.11 ± 2.51 | 64.45 ± 13.35 | 7/10 | 1/7 |
| CIFAR-100 | Fixed temperatures | 83.72 ± 2.24 | 70.93 ± 10.56 | 10/10 | 5/10 |
| CIFAR-100 | Tip-Adapter sum | 83.21 ± 1.71 | 65.36 ± 12.68 | 5/10 | 0/5 |
| CIFAR-100 | Tip-Adapter mean | 83.19 ± 1.72 | 68.69 ± 10.39 | 9/10 | 2/9 |
| Pets | Retained references | 89.95 ± 2.67 | 93.79 ± 3.83 | 10/10 | 1/10 |
| Pets | Fixed-gate removal | 90.43 ± 3.16 | 93.81 ± 3.11 | 8/10 | 3/8 |
| Pets | Recalibrated removal | 90.48 ± 3.04 | 93.44 ± 3.56 | 8/10 | 3/8 |
| Pets | Dense text + prototypes | 89.46 ± 3.08 | 94.47 ± 2.60 | 10/10 | 2/10 |
| Pets | Raw-score fusion | 91.35 ± 2.91 | 92.72 ± 2.70 | 9/10 | 4/9 |
| Pets | Fixed temperatures | 90.48 ± 3.42 | 94.21 ± 3.19 | 9/10 | 3/9 |
| Pets | Tip-Adapter sum | 89.18 ± 1.73 | 91.61 ± 6.00 | 7/10 | 1/7 |
| Pets | Tip-Adapter mean | 90.73 ± 2.17 | 91.39 ± 4.32 | 10/10 | 4/10 |
| ImageNet-R | Retained references | 86.15 ± 0.94 | 84.63 ± 0.95 | 10/10 | 2/10 |
| ImageNet-R | Fixed-gate removal | 86.17 ± 0.65 | 66.89 ± 22.80 | 10/10 | 3/10 |
| ImageNet-R | Recalibrated removal | 86.16 ± 0.71 | 66.88 ± 22.79 | 10/10 | 3/10 |
| ImageNet-R | Dense text + prototypes | 86.01 ± 0.94 | 83.94 ± 1.10 | 10/10 | 3/10 |
| ImageNet-R | Raw-score fusion | 86.15 ± 0.68 | 65.55 ± 20.42 | 10/10 | 2/10 |
| ImageNet-R | Fixed temperatures | 85.99 ± 0.71 | 82.92 ± 1.03 | 10/10 | 1/10 |
| ImageNet-R | Tip-Adapter sum | 86.06 ± 0.78 | 74.52 ± 3.69 | 10/10 | 2/10 |
| ImageNet-R | Tip-Adapter mean | 86.08 ± 1.08 | 81.61 ± 1.51 | 10/10 | 4/10 |

Table S20: ViT-L/14 matched-test 95% retention frontier across ten class partitions. Values are percentages, mean ± sample standard deviation over feasible partitions after averaging orders. Test met equals feasibility by construction in this descriptive sweep.

| Data | Readout | Base | Novel | Feasible | Test met |
|---|---|---|---|---|---|
| CIFAR-100 | Retained references | 83.91 ± 1.42 | 78.02 ± 2.27 | 10/10 | 10/10 |
| CIFAR-100 | Fixed-gate removal | 83.25 ± 1.17 | 61.68 ± 9.35 | 6/10 | 6/6 |
| CIFAR-100 | Recalibrated removal | 83.25 ± 1.17 | 63.78 ± 6.52 | 6/10 | 6/6 |
| CIFAR-100 | Dense text + prototypes | 83.91 ± 1.42 | 76.18 ± 2.56 | 10/10 | 10/10 |
| CIFAR-100 | Raw-score fusion | 83.25 ± 1.17 | 62.37 ± 6.38 | 6/10 | 6/6 |
| CIFAR-100 | Fixed temperatures | 83.64 ± 1.21 | 74.53 ± 2.70 | 9/10 | 9/9 |
| CIFAR-100 | Tip-Adapter sum | 83.16 ± 1.28 | 61.83 ± 4.92 | 5/10 | 5/5 |
| CIFAR-100 | Tip-Adapter mean | 83.91 ± 1.42 | 65.20 ± 11.69 | 10/10 | 10/10 |
| Pets | Retained references | 91.56 ± 1.87 | 93.30 ± 3.74 | 10/10 | 10/10 |
| Pets | Fixed-gate removal | 91.57 ± 1.88 | 93.78 ± 2.17 | 10/10 | 10/10 |
| Pets | Recalibrated removal | 91.56 ± 1.88 | 93.32 ± 2.10 | 10/10 | 10/10 |
| Pets | Dense text + prototypes | 91.56 ± 1.87 | 93.79 ± 2.78 | 10/10 | 10/10 |
| Pets | Raw-score fusion | 91.57 ± 1.88 | 92.39 ± 2.75 | 10/10 | 10/10 |
| Pets | Fixed temperatures | 91.56 ± 1.87 | 93.64 ± 2.48 | 10/10 | 10/10 |
| Pets | Tip-Adapter sum | 91.51 ± 1.98 | 87.78 ± 9.98 | 9/10 | 9/9 |
| Pets | Tip-Adapter mean | 91.57 ± 1.87 | 92.02 ± 4.15 | 10/10 | 10/10 |
| ImageNet-R | Retained references | 86.42 ± 0.53 | 84.31 ± 1.01 | 10/10 | 10/10 |
| ImageNet-R | Fixed-gate removal | 86.41 ± 0.56 | 73.82 ± 6.70 | 9/10 | 9/9 |
| ImageNet-R | Recalibrated removal | 86.42 ± 0.56 | 73.50 ± 6.79 | 9/10 | 9/9 |
| ImageNet-R | Dense text + prototypes | 86.42 ± 0.53 | 83.02 ± 1.40 | 10/10 | 10/10 |
| ImageNet-R | Raw-score fusion | 86.42 ± 0.56 | 70.85 ± 8.03 | 9/10 | 9/9 |
| ImageNet-R | Fixed temperatures | 86.43 ± 0.53 | 82.49 ± 1.11 | 10/10 | 10/10 |
| ImageNet-R | Tip-Adapter sum | 86.43 ± 0.56 | 73.16 ± 5.66 | 9/10 | 9/9 |
| ImageNet-R | Tip-Adapter mean | 86.43 ± 0.52 | 80.91 ± 1.48 | 10/10 | 10/10 |

Table S21: ViT-B/32 matched-test 95% frontier under the same access, aggregation and feasibility reporting as Table S20. Absolute targets are encoder-specific.

| Data | Readout | Base | Novel | Feasible | Test met |
|---|---|---|---|---|---|
| CIFAR-100 | Retained references | 76.16 ± 1.99 | 59.64 ± 2.10 | 10/10 | 10/10 |
| CIFAR-100 | Fixed-gate removal | 74.84 ± 1.75 | 49.81 ± 10.48 | 4/10 | 4/4 |
| CIFAR-100 | Recalibrated removal | 74.84 ± 1.75 | 49.81 ± 10.48 | 4/10 | 4/4 |
| CIFAR-100 | Dense text + prototypes | 76.15 ± 2.22 | 51.78 ± 8.42 | 8/10 | 8/8 |
| CIFAR-100 | Raw-score fusion | 75.12 ± 1.63 | 46.30 ± 9.93 | 5/10 | 5/5 |
| CIFAR-100 | Fixed temperatures | 75.71 ± 1.46 | 54.14 ± 4.58 | 9/10 | 9/9 |
| CIFAR-100 | Tip-Adapter sum | 74.08 ± 1.05 | 52.81 ± 5.29 | 3/10 | 3/3 |
| CIFAR-100 | Tip-Adapter mean | 76.16 ± 2.11 | 46.50 ± 11.77 | 9/10 | 9/9 |
| Pets | Retained references | 86.92 ± 2.17 | 76.54 ± 6.71 | 10/10 | 10/10 |
| Pets | Fixed-gate removal | 86.45 ± 2.27 | 77.62 ± 6.10 | 5/10 | 5/5 |
| Pets | Recalibrated removal | 86.46 ± 2.29 | 78.61 ± 5.07 | 5/10 | 5/5 |
| Pets | Dense text + prototypes | 86.84 ± 2.33 | 71.48 ± 15.80 | 8/10 | 8/8 |
| Pets | Raw-score fusion | 87.12 ± 2.19 | 67.25 ± 11.41 | 9/10 | 9/9 |
| Pets | Fixed temperatures | 86.93 ± 2.16 | 74.97 ± 6.20 | 10/10 | 10/10 |
| Pets | Tip-Adapter sum | 86.83 ± 2.22 | 70.08 ± 9.72 | 6/10 | 6/6 |
| Pets | Tip-Adapter mean | 87.28 ± 2.29 | 68.64 ± 21.10 | 8/10 | 8/8 |
| ImageNet-R | Retained references | 70.38 ± 0.87 | 56.36 ± 1.11 | 10/10 | 10/10 |
| ImageNet-R | Fixed-gate removal | 69.72 | 31.36 | 1/10 | 1/1 |
| ImageNet-R | Recalibrated removal | 69.71 | 30.54 | 1/10 | 1/1 |
| ImageNet-R | Dense text + prototypes | 70.37 ± 0.88 | 48.30 ± 3.45 | 10/10 | 10/10 |
| ImageNet-R | Raw-score fusion | 69.71 | 28.06 | 1/10 | 1/1 |
| ImageNet-R | Fixed temperatures | 70.38 ± 0.87 | 53.72 ± 1.35 | 10/10 | 10/10 |
| ImageNet-R | Tip-Adapter sum | 69.67 ± 0.03 | 35.63 ± 8.58 | 2/10 | 2/2 |
| ImageNet-R | Tip-Adapter mean | 70.45 ± 0.92 | 41.88 ± 7.58 | 8/10 | 8/8 |

Table S22: ViT-B/32 validation-selected 95% policy. Values are achieved base/novel test percentages over validation-feasible partitions; Test met counts those meeting the independent test-retention criterion. Validation feasibility does not guarantee test target attainment.

| Data | Readout | Base | Novel | Feasible | Test met |
|---|---|---|---|---|---|
| CIFAR-100 | Retained references | 75.98 ± 2.43 | 59.04 ± 3.66 | 10/10 | 5/10 |
| CIFAR-100 | Fixed-gate removal | 73.46 ± 1.73 | 52.54 ± 10.71 | 7/10 | 1/7 |
| CIFAR-100 | Recalibrated removal | 73.40 ± 1.79 | 54.20 ± 6.94 | 7/10 | 1/7 |
| CIFAR-100 | Dense text + prototypes | 74.30 ± 2.85 | 55.73 ± 8.03 | 7/10 | 2/7 |
| CIFAR-100 | Raw-score fusion | 73.82 ± 2.67 | 49.47 ± 12.37 | 8/10 | 2/8 |
| CIFAR-100 | Fixed temperatures | 73.90 ± 2.39 | 56.16 ± 8.19 | 9/10 | 1/9 |
| CIFAR-100 | Tip-Adapter sum | 73.55 ± 2.53 | 49.25 ± 12.78 | 7/10 | 1/7 |
| CIFAR-100 | Tip-Adapter mean | 73.54 ± 2.88 | 57.47 ± 8.46 | 7/10 | 0/7 |
| Pets | Retained references | 88.16 ± 3.34 | 72.75 ± 8.65 | 10/10 | 8/10 |
| Pets | Fixed-gate removal | 86.17 ± 2.74 | 76.30 ± 6.46 | 5/10 | 3/5 |
| Pets | Recalibrated removal | 86.24 ± 2.14 | 77.70 ± 5.37 | 5/10 | 2/5 |
| Pets | Dense text + prototypes | 86.89 ± 3.09 | 70.27 ± 15.78 | 7/10 | 3/7 |
| Pets | Raw-score fusion | 86.93 ± 5.36 | 63.79 ± 15.01 | 9/10 | 5/9 |
| Pets | Fixed temperatures | 85.53 ± 5.27 | 74.57 ± 7.40 | 10/10 | 4/10 |
| Pets | Tip-Adapter sum | 86.87 ± 2.42 | 73.74 ± 4.16 | 4/10 | 2/4 |
| Pets | Tip-Adapter mean | 84.97 ± 3.13 | 81.94 ± 3.51 | 5/10 | 1/5 |
| ImageNet-R | Retained references | 70.18 ± 0.96 | 56.80 ± 1.33 | 10/10 | 3/10 |
| ImageNet-R | Fixed-gate removal | 68.23 ± 0.84 | 19.12 ± 6.82 | 2/10 | 0/2 |
| ImageNet-R | Recalibrated removal | 68.23 ± 0.84 | 19.12 ± 6.82 | 2/10 | 0/2 |
| ImageNet-R | Dense text + prototypes | 70.43 ± 1.50 | 46.95 ± 4.97 | 10/10 | 6/10 |
| ImageNet-R | Raw-score fusion | 68.16 ± 0.84 | 21.79 ± 6.87 | 2/10 | 0/2 |
| ImageNet-R | Fixed temperatures | 70.46 ± 1.26 | 53.38 ± 1.43 | 10/10 | 6/10 |
| ImageNet-R | Tip-Adapter sum | 68.89 ± 0.79 | 38.12 ± 2.27 | 3/10 | 0/3 |
| ImageNet-R | Tip-Adapter mean | 69.99 ± 1.61 | 46.39 ± 3.62 | 9/10 | 4/9 |

## Appendix J. Independent protection-risk calibration

### *Appendix J.1. Disjoint access and refitting*

For each dataset, original fitting and validation features are concatenated in that order. A generator with seed 662026 shuffles each class, reserving $\max(1, \lfloor 0.4N_c \rfloor)$ calibration and $\max(1, \lfloor 0.2N_c \rfloor)$ tuning observations; the remainder fits the readouts. Every class retains at least two fitting images. Identical indices are used by both encoders and every method. The global fitting/tuning/calibration sizes are 4,437/2,139/4,318 on CIFAR, 556/248/510 on Pets, and 5,096/2,390/4,863 on ImageNet-R. Each true partition uses only its base classes in these pools. Support and test observations remain outside all three. All ridge parameters, pseudo-episode mixtures, branch temperatures and semantic-cache hyperparameters are recomputed using fitting and tuning data exclusively.

Only frozen features are reused; previously tuned score models are not treated as independent of their old validation labels. The new calibration labels enter only threshold certification or the explicitly compared empirical threshold controls. The independent-pool experiment is a benchmark revision on previously studied test data, not a newly untouched final evaluation. The statistical analysis assumes that, conditional on each base class and the fitted models, calibration observations are independent draws from the class-conditional distribution to which the population statement applies. Dataset shift and adaptive support arrivals depending on calibration labels are outside that statement.

*Appendix J.2. Weighted protection bound*

Let $B$ be the number of base classes and $n_c$ their calibration counts. Assign each observation of class $c$ weight $w_i = 1/(Bn_c)$, so $\sum_i w_i = 1$, and let $b = \max_i w_i$. For any fixed bounded losses $L_i \in [0, 1]$, write $q = \sum_i w_i L_i$ and $\mu = \sum_i w_i \mathbb{E}L_i$. Conditional on the score models and the independently selected threshold family, independence gives, for $t < 0$,

$$\log \mathbb{E}e^{tq} \le \sum_i \log(1 - \mu_i + \mu_i e^{tw_i}) \tag{J.1}$$

$$\le b^{-1} \sum_i w_i \log(1 - \mu_i + \mu_i e^{tb}) \tag{J.2}$$

$$\le b^{-1} \log(1 - \mu + \mu e^{tb}), \tag{J.3}$$

where $\mu_i = \mathbb{E}L_i$. The first inequality uses boundedness and the chord of the exponential; the second uses convexity of each log-moment-generating function in $w_i$ and its zero value at zero; the third uses concavity in $\mu_i$. Chernoff optimization yields

$$\Pr(q \le x) \le \exp[-\operatorname{kl}(x, \mu)/b], \qquad x < \mu, \tag{J.4}$$

where kl is Bernoulli relative entropy. Thus, for the null $\mu \ge \alpha$, the value

$$p(q) = \begin{cases} \exp[-\operatorname{kl}(q, \alpha)/b], & q < \alpha, \\ 1, & q \ge \alpha \end{cases} \tag{J.5}$$

is a valid lower-tail p-value. Unlike an ordinary pooled binomial calculation, this bound retains the unequal macro weights. Its dependence on the largest weight makes sparse class-conditional calibration sets conservative.

Apply the bound to the protection loss of main Eq. (15). Let $A_{\mathrm{ref}} = \Pr(R = 1)$, $D = \Pr(R = 1, C_d = 0)$ and $G = \Pr(R = 0, C_d = 1)$. Then

$$\mathbb{E}L_d = D + (1 - r)(1 - A_{\mathrm{ref}}), \qquad A_B(d) = A_{\mathrm{ref}} - D + G. \tag{J.6}$$

Consequently $\mathbb{E}L_d \leq 1 - r$ implies $A_B(d) \geq rA_{\mathrm{ref}}$. The reference uncertainty is included in the paired loss rather than replaced by a point estimate. Retained base scores have $G = 0$; other readouts may have $G > 0$, so protection is sufficient and can be stricter than accuracy-ratio control for replacement models.

*Appendix J.3. Threshold search and trajectory confidence*

For each method and arrival, the independently fixed finite threshold family consists of unique quantiles at 0,1,. . . ,100% of base tuning margins, in increasing order of novel admission. Following fixed-sequence testing [21], test this family in order at level $\delta/5$, stop at the first failure, and return the last accepted threshold. A false certification requires rejecting the first true null in the fixed order, whose probability is at most $\delta/5$; independence between these tests is unnecessary. A union bound over the five prespecified arrivals gives error probability at most $\delta$ for each method/order/partition. The same calibration pool may be reused because the score models, threshold families and support arrivals do not adapt to its labels. No simultaneous confidence claim across methods, orders, partitions, datasets or separately reported $\delta$ values is made.

If the accepted finite set is empty, the policy returns the original reference classifier and admits no novel class. This fallback structurally preserves its original accuracy and has zero novel accuracy. It is reported separately from

useful certified expansion. The confidence levels 90,95,99% and empirical target fractions 95,97,99% were specified before the new experiment; the primary setting uses 95% retention and 95% confidence. Their complete results, worst shortfalls and five-session trajectories are saved in `independen t_risk_summary.json` and `independent_risk_summary.csv`.

Table S23: Complete ViT-L/14 independent-calibration comparison, with meanings and fallback accounting as in main Table 2. All operational means include every partition; certification counts identify finite accepted thresholds.

| Dataset | Readout | Emp. met | Emp. novel | Cert. exp. | Test met | Cert. base | Cert. novel |
|---|---|---|---|---|---|---|---|
| CIFAR-100 | Retained | 7/10 | 76.28 | 0/10 | 10/10 | 88.29 | 0.00 |
| CIFAR-100 | Recalibrated removal | 6/10 | 35.32 | 0/10 | 10/10 | 88.29 | 0.00 |
| CIFAR-100 | Dense prototypes | 6/10 | 71.92 | 0/10 | 10/10 | 88.29 | 0.00 |
| CIFAR-100 | Raw fusion | 5/10 | 34.46 | 0/10 | 10/10 | 88.29 | 0.00 |
| CIFAR-100 | Fixed temperature | 8/10 | 71.36 | 0/10 | 10/10 | 88.29 | 0.00 |
| CIFAR-100 | Class-mean cache | 4/10 | 65.73 | 0/10 | 10/10 | 88.29 | 0.00 |
| Pets | Retained | 1/10 | 93.07 | 0/10 | 10/10 | 96.05 | 0.00 |
| Pets | Recalibrated removal | 4/10 | 93.29 | 0/10 | 10/10 | 96.05 | 0.00 |
| Pets | Dense prototypes | 0/10 | 95.08 | 0/10 | 10/10 | 96.05 | 0.00 |
| Pets | Raw fusion | 4/10 | 86.90 | 0/10 | 10/10 | 96.05 | 0.00 |
| Pets | Fixed temperature | 3/10 | 91.87 | 0/10 | 10/10 | 96.05 | 0.00 |
| Pets | Class-mean cache | 1/10 | 92.50 | 0/10 | 10/10 | 96.05 | 0.00 |
| ImageNet-R | Retained | 7/10 | 83.45 | 10/10 | 10/10 | 89.73 | 77.64 |
| ImageNet-R | Recalibrated removal | 4/10 | 57.36 | 0/10 | 10/10 | 91.54 | 0.00 |
| ImageNet-R | Dense prototypes | 6/10 | 82.47 | 5/10 | 10/10 | 91.04 | 24.12 |
| ImageNet-R | Raw fusion | 3/10 | 64.80 | 4/10 | 10/10 | 91.08 | 26.42 |
| ImageNet-R | Fixed temperature | 5/10 | 81.28 | 10/10 | 10/10 | 89.76 | 65.69 |
| ImageNet-R | Class-mean cache | 4/10 | 79.37 | 2/10 | 10/10 | 91.18 | 6.20 |

Table S24: Complete ViT-B/32 independent-calibration comparison. Infeasible certification triggers the shared base-only reference fallback and zero novel accuracy.

| Dataset | Readout | Emp. met | Emp. novel | Cert. exp. | Test met | Cert. base | Cert. novel |
|---|---|---|---|---|---|---|---|
| CIFAR-100 | Retained | 1/10 | 60.64 | 0/10 | 10/10 | 79.08 | 0.00 |
| CIFAR-100 | Recalibrated removal | 2/10 | 47.79 | 0/10 | 10/10 | 79.08 | 0.00 |
| CIFAR-100 | Dense prototypes | 2/10 | 52.74 | 0/10 | 10/10 | 79.08 | 0.00 |
| CIFAR-100 | Raw fusion | 2/10 | 50.33 | 0/10 | 10/10 | 79.08 | 0.00 |
| CIFAR-100 | Fixed temperature | 5/10 | 55.06 | 0/10 | 10/10 | 79.08 | 0.00 |
| CIFAR-100 | Class-mean cache | 2/10 | 57.79 | 0/10 | 10/10 | 79.08 | 0.00 |
| Pets | Retained | 1/10 | 75.10 | 0/10 | 10/10 | 90.24 | 0.00 |
| Pets | Recalibrated removal | 10/10 | 28.49 | 0/10 | 10/10 | 90.24 | 0.00 |
| Pets | Dense prototypes | 4/10 | 54.67 | 0/10 | 10/10 | 90.24 | 0.00 |
| Pets | Raw fusion | 6/10 | 53.57 | 0/10 | 10/10 | 90.24 | 0.00 |
| Pets | Fixed temperature | 5/10 | 64.90 | 0/10 | 10/10 | 90.24 | 0.00 |
| Pets | Class-mean cache | 7/10 | 42.00 | 0/10 | 10/10 | 90.24 | 0.00 |
| ImageNet-R | Retained | 2/10 | 56.51 | 10/10 | 10/10 | 73.66 | 45.30 |
| ImageNet-R | Recalibrated removal | 9/10 | 3.58 | 0/10 | 10/10 | 75.01 | 0.00 |
| ImageNet-R | Dense prototypes | 4/10 | 43.48 | 0/10 | 10/10 | 75.01 | 0.00 |
| ImageNet-R | Raw fusion | 9/10 | 3.75 | 0/10 | 10/10 | 75.01 | 0.00 |
| ImageNet-R | Fixed temperature | 2/10 | 52.45 | 7/10 | 10/10 | 74.24 | 27.28 |
| ImageNet-R | Class-mean cache | 5/10 | 33.38 | 0/10 | 10/10 | 75.01 | 0.00 |

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